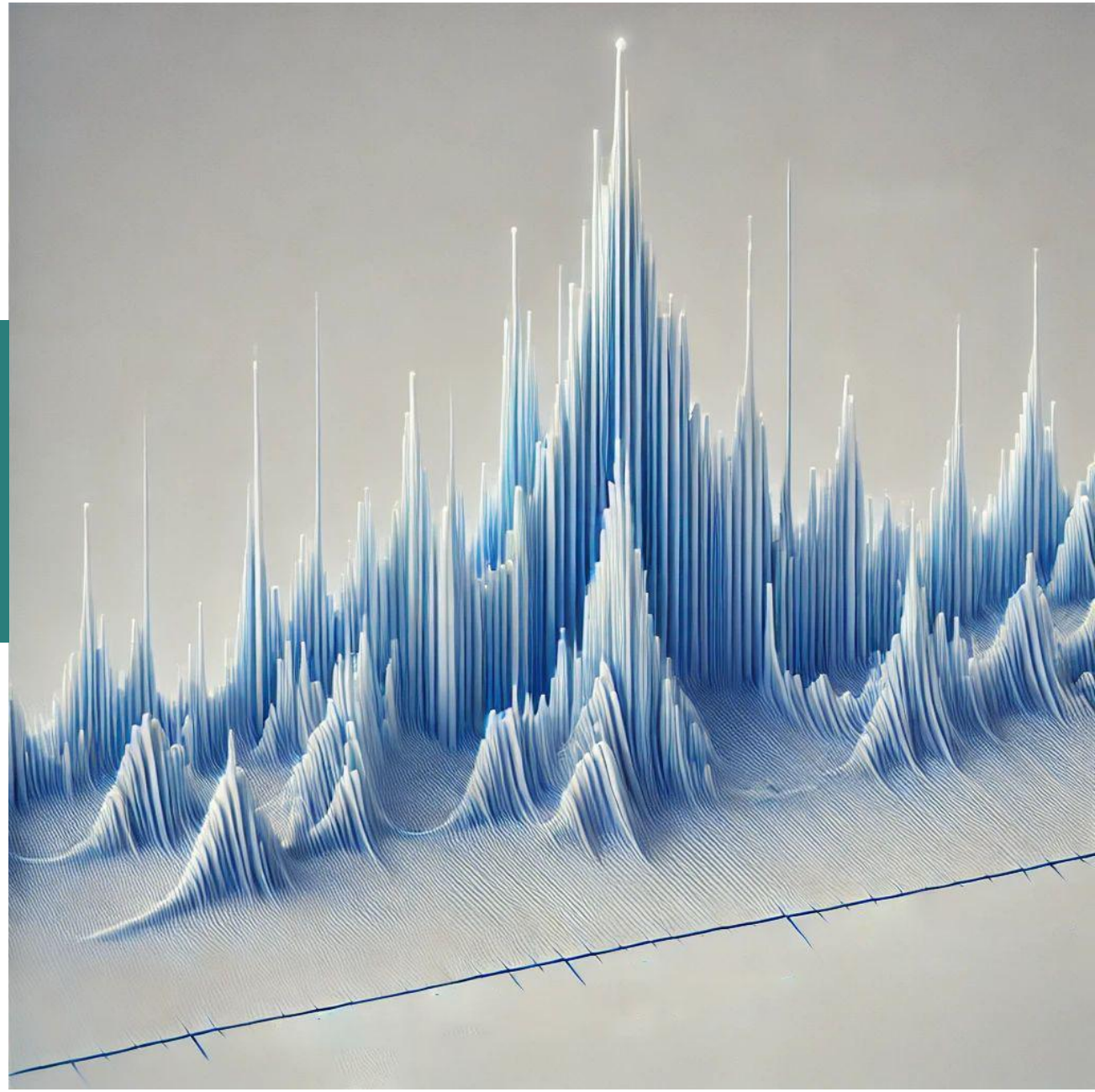
Rietveld Refinement of Powder Diffraction Data



ACerS Workshop: Tools for Visualizing and Understanding the Structure of Crystalline Ceramics

Taylor Sparks
Professor of Materials Science & Engineering, University of Utah

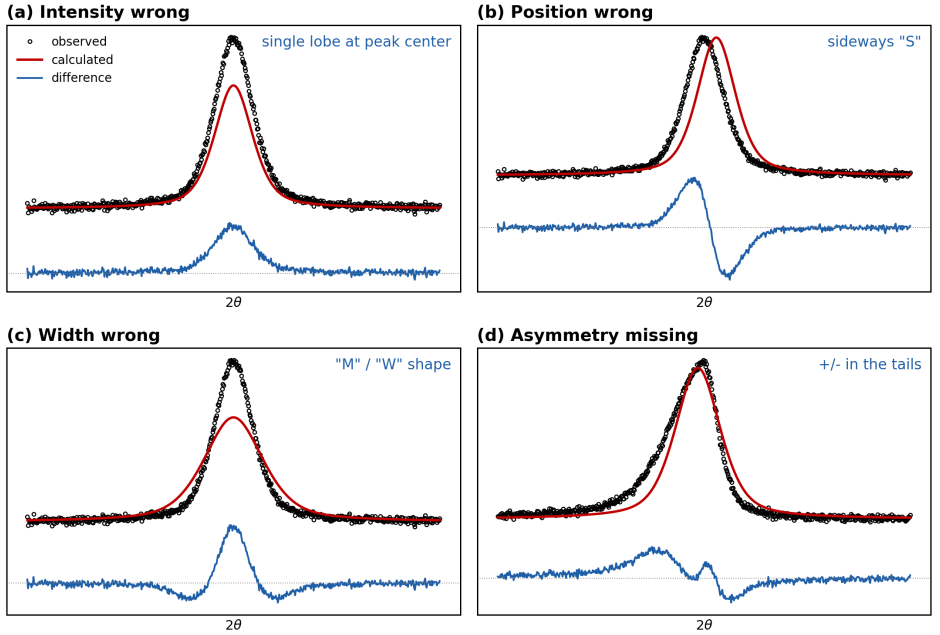
Download the software and tutorial materials now, the files are large

GSAS-II software
~400 MB download, 2 GB installed

Workshop tutorial materials
~30 MB

One technique answers many quantitative questions

- What is Rietveld refinement and how does it work?
- Quantitative analysis of multiphase mixtures
  - Amorphous content
- Atomic positions and site occupancy
- Thermal displacement parameters
- Preferred orientation
- Crystallite size and strain



Rietveld refinement least-squares fits a calculated pattern to your data

- **Hugo Rietveld (1969)**
- Calculated pattern vs measured pattern
- Variable model parameters, adjusted by least squares
- Accounts for measurement errors (shift, zero offset)
- Different from Le Bail or Pawley (profile) fitting: those fit intensities without a structure
- Ab initio structure determination is a separate, harder problem

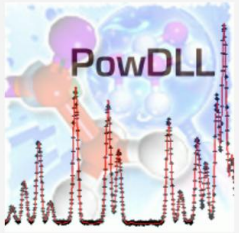


GSAS-II and PowDLL cover the whole workflow for free

PowDLL .NET converter for XRPD files

Home (Lab) Download Hints'n'Tips Screenshots For programmers Updates Contact

PowDLL is a .NET dynamic link library used for the interconversion procedure between variable formats of Powder X-Ray files. The DLL is capable of handling the most common file formats (binary and ASCII). The library can be used as a reusable component with any .NET language or as a **standalone utility**.



PowDLL can run on windows OS as long as they have dotnet runtimes version 2 (or later) installed (i.e., Windows Vista SP1 or later). Linux is also supported through Wine.

Imports

Bruker/Siemens RAW (versions 1,2,4), Bruker BRML, STOE RAW (plus multi-range files), Scintag RAW (plus multi-range files), Rigaku RAW, Shimadzu RAW, Philips RD, Philips SD, Scintag RD, Panalytical XRDML, INEL Binary, INEL ASCII, Scintag ARD, powderCIF, Sietronics CPI, Riet7 DAT, DBWS, GSAS (CW STD), Jade MDI, Rigaku RIG, Philips UDF, UXD, XDA, XDD, CCDC Mercury XYE, XPOWDER PLV (old and new format), UDF (NEX), ProtoXRD and ASCII XY Files.

Exports

Bruker/Siemens RAW (versions 1,2), Philips RD, Scintag ARD, Sietronics CPI, Riet7 DAT, DBWS, GSAS (CW STD), Jade MDI, Rigaku RIG, Philips UDF, UXD, XDA, XDD, Panalytical XRDML, ASCII XY Files, MS-Excel Multiple XY, Xpowder PLV files.

Citation

Please cite if you find PowDLL useful:

PowDLL, a reusable .NET component for interconverting powder diffraction data: Recent developments, N. Kourkoumelis, ICDD Annual Spring Meetings (ed. Lisa O'Neill), Powder Diffraction, 28 (2013) 137-48.

GSAS-II Home Page

GSAS-II

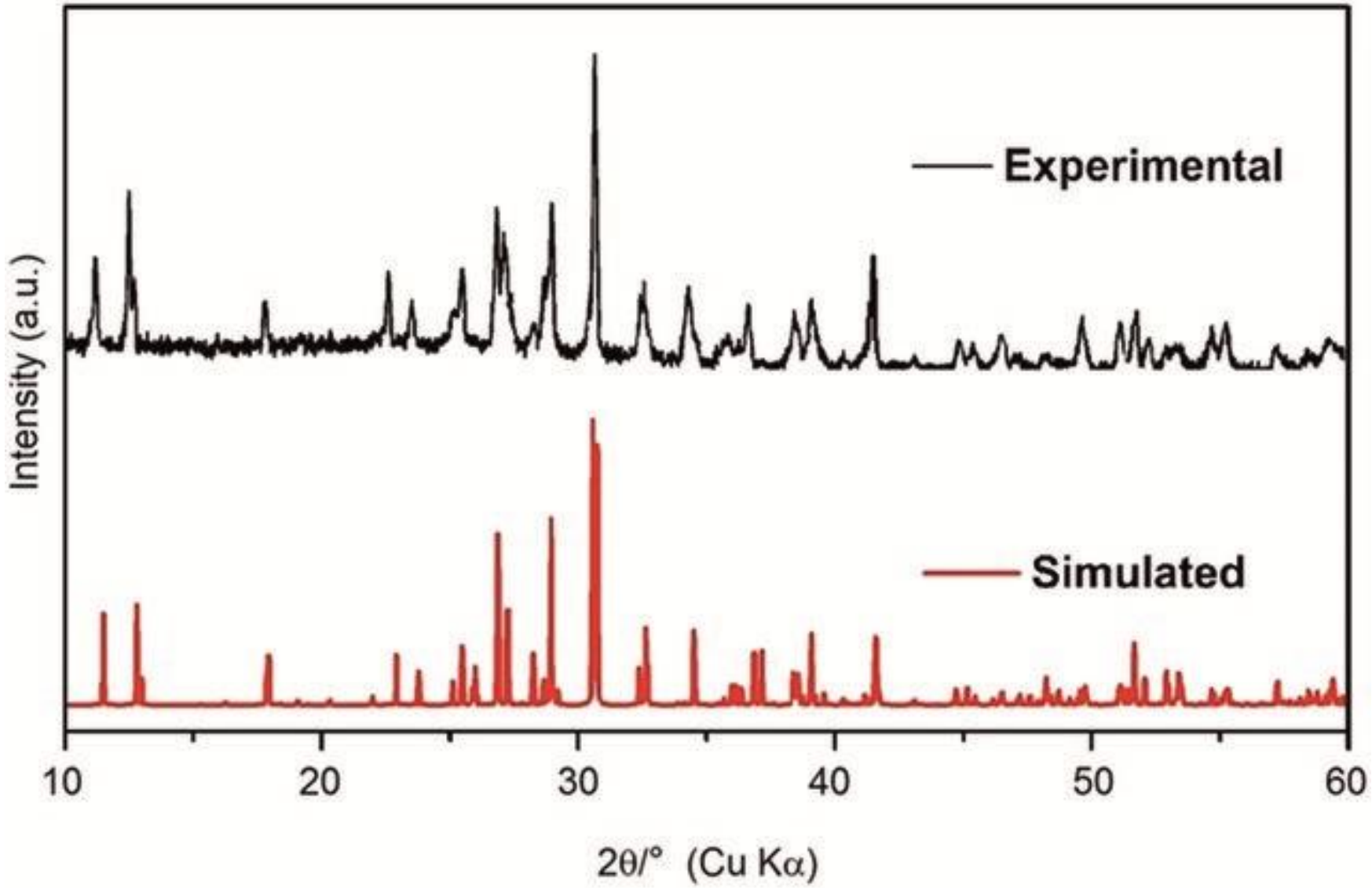
Contents

1. GSAS-II Installation Options
2. Available Documentation for GSAS-II
3. Mailing List
4. Reporting GSAS-II Bugs
5. Information for GSAS-II Developers
6. Compiling GSAS-II
7. Customized Python Installation
8. Web Proxies

Welcome to the home page for GSAS-II, a unique and comprehensive open source Python project for determination of crystal structures and diffraction-based materials characterization for crystalline solids on all scales, from perovskites through proteins, using both powder and single-crystal diffraction and with both x-ray and neutron probes. Refinements can combine measurements from laboratory and synchrotron x-rays, as well as constant wavelength or time-of-flight neutron sources. It provides structure solution and refinement, as well as extensive visualization capabilities.

GSAS-II is made available for free use (see license) with open access to the source code.

Modern software simulates powder patterns remarkably well

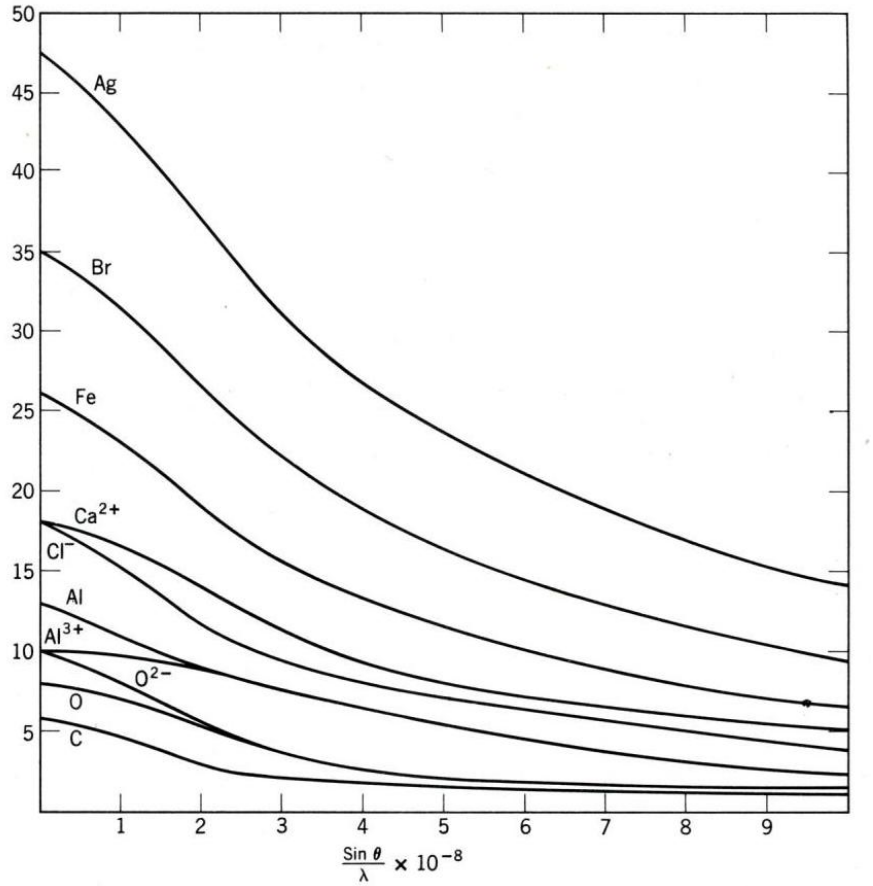


Peak intensity folds in source, sample, and structure factors

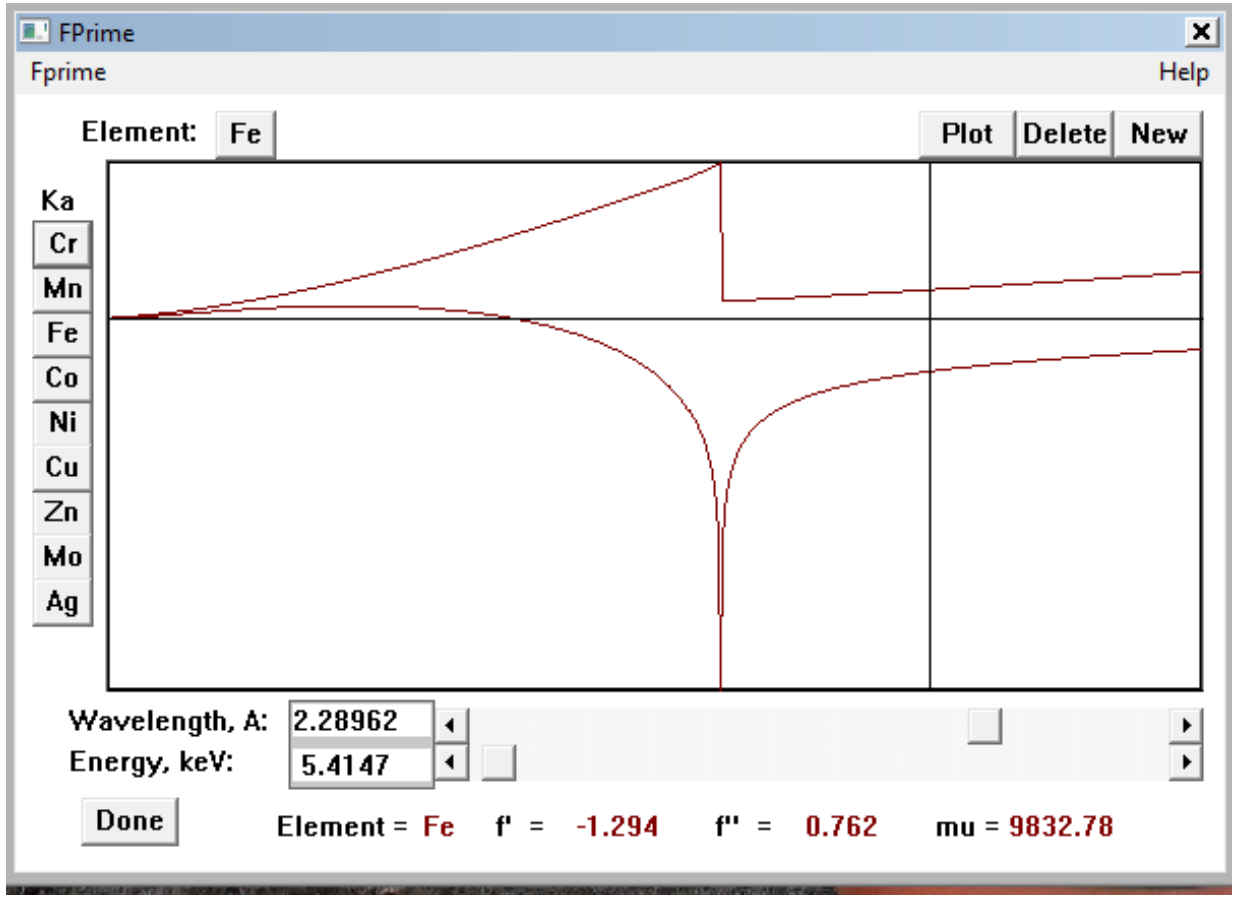
$$I_{(hkl)\alpha} = \frac{I_0 \lambda^3}{64\pi r}\left(\frac{e^2}{m_e c^2}\right)^2 \frac{M_{(hkl)}}{V_\alpha^2}\left|F_{(hkl)\alpha}\right|^2 \left(\frac{1+\cos^2(2\theta)\cos^2(2\theta_m)}{\sin^2\theta\cos\theta}\right)_{hkl} \frac{v_\alpha}{\mu_s}$$

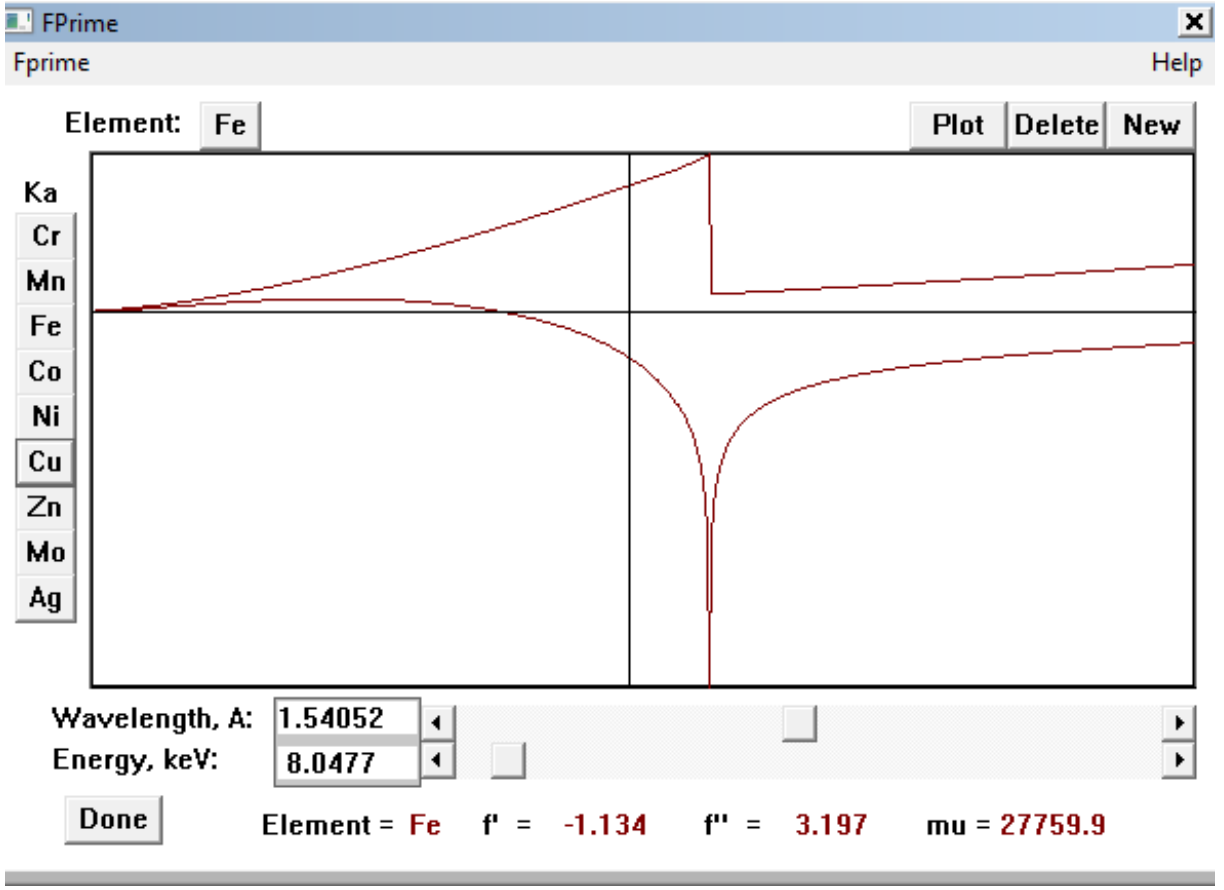
$$F_{hkl} = \sum_{j=1}^{m} N_j f_j \exp\left[2\pi i\left(hx_j + ky_j + lz_j\right)\right]$$

$$|f|^2 = \left(f_0 \exp\left[\frac{-B\sin^2\theta}{\lambda^2}\right] + \Delta f'\right)^2 + (\Delta f'')^2$$

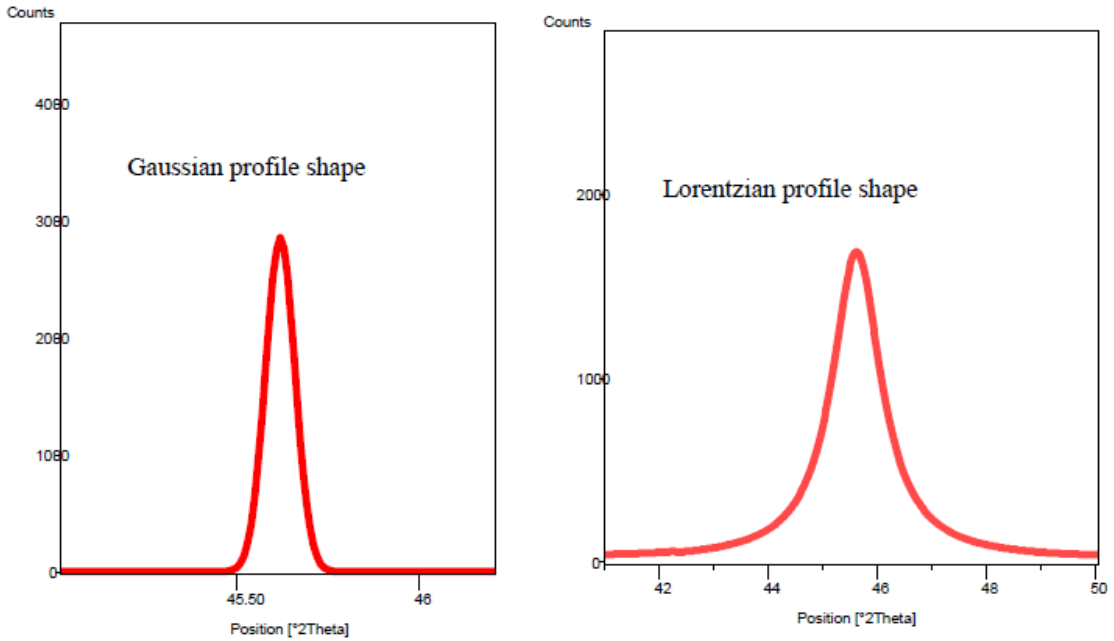


Radiation choice sets the anomalous scattering factors f' and f''





The instrument dictates Gaussian vs Lorentzian peak character



$$H_k = \left(U \tan^2 \theta + V \tan \theta + W\right)^{1/2}$$

prism.mit.edu/xray/Rietveld_XRDSimulation.pdf

Every parameter changes the pattern in one of three ways

WHERE peaks sit

Lattice parameters (a, b, c, angles)

Zero offset

Sample displacement

WHAT SHAPE they have

Gaussian widths: U, V, W

Lorentzian widths: X, Y

Asymmetry (axial divergence)

Size and strain broadening

HOW TALL they are

Scale factor(s)

Atomic positions (x, y, z)

Site occupancies

Thermal parameters (Uiso, Uij)

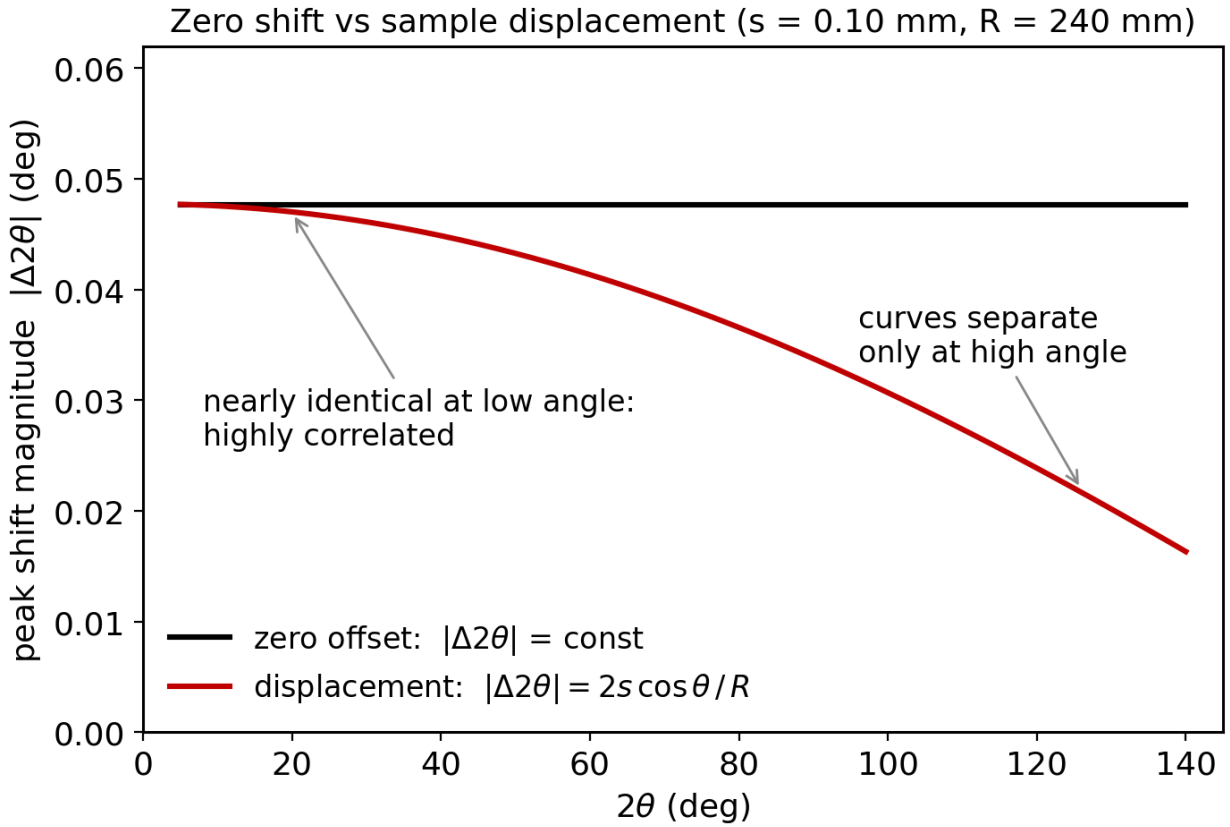
Preferred orientation

Absorption, roughness

Diagnosis strategy: when the fit is bad, ask which bucket is wrong. Each bucket leaves a different fingerprint in the difference curve.

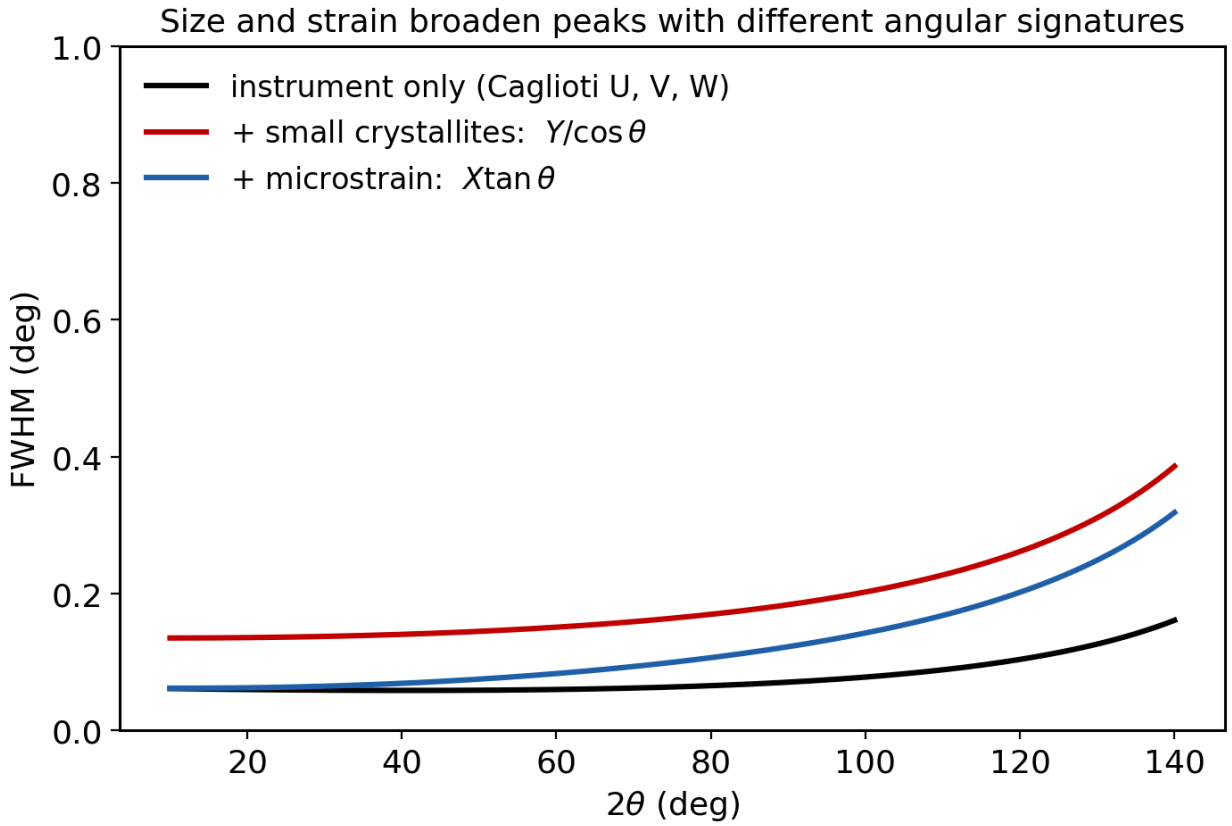
Peak positions must match before anything else can refine

- **Two rival corrections:**
  - Zero offset: Δ2θ = const.
  - Displacement: Δ2θ = −2s cosθ/R
- Nearly identical at low angle, so the two are highly correlated
- Fix: internal standard (NIST Si 640) pins one of them
- If positions are off, nothing downstream can converge (McCusker et al., 1999)



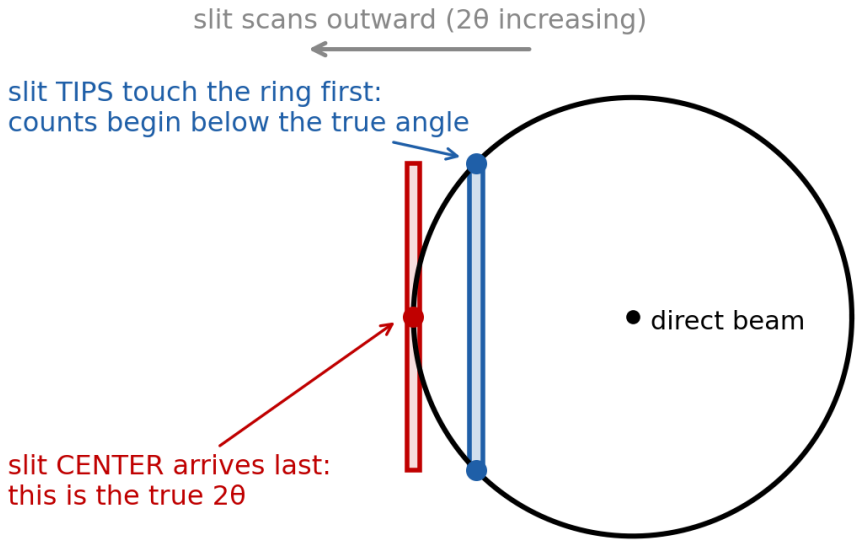
Peak widths encode real microstructure through U, V, W, X, Y

- **Gaussian part (Caglioti, 1958):**
  - $FWHM^2 = U \tan^2\theta + V \tan\theta + W$
- **Lorentzian part:**
  - $FWHM = X \tan\theta + Y / \cos\theta$
- X grows like strain broadening ($\tan\theta$)
- Y grows like size broadening ($1/\cos\theta$, Scherrer)
- Profile terms are physics, not fudge factors



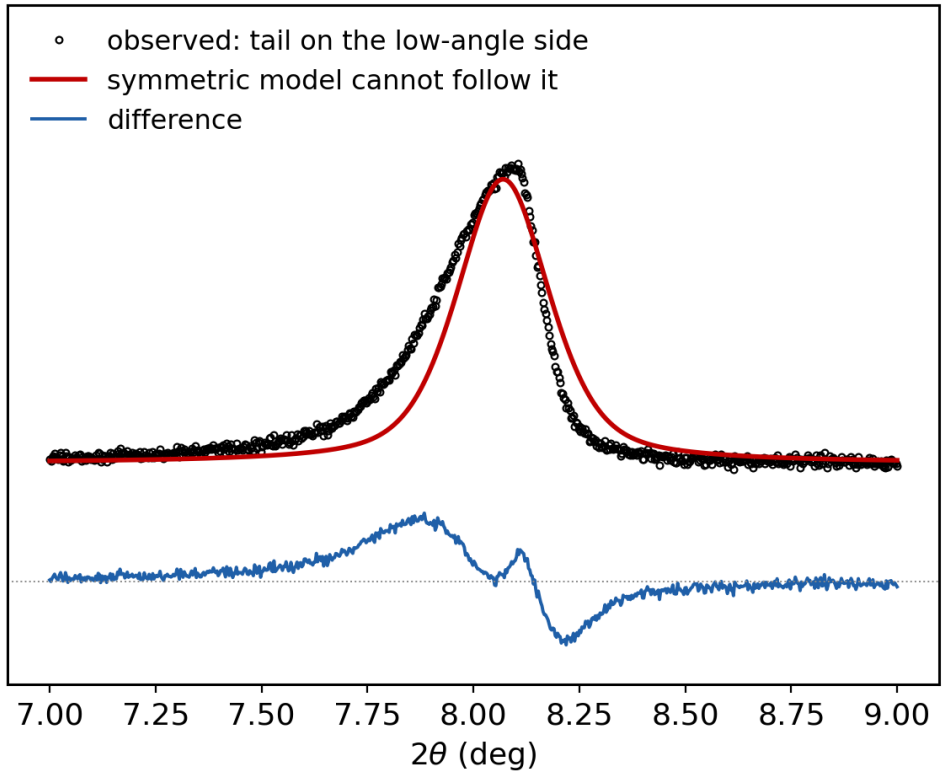
Axial divergence makes low-angle peaks lean; correct it, don't absorb it

Detector view: a straight slit meets a curved ring



early shift ≈ H²/2r: small ring (low 2θ) = big shift

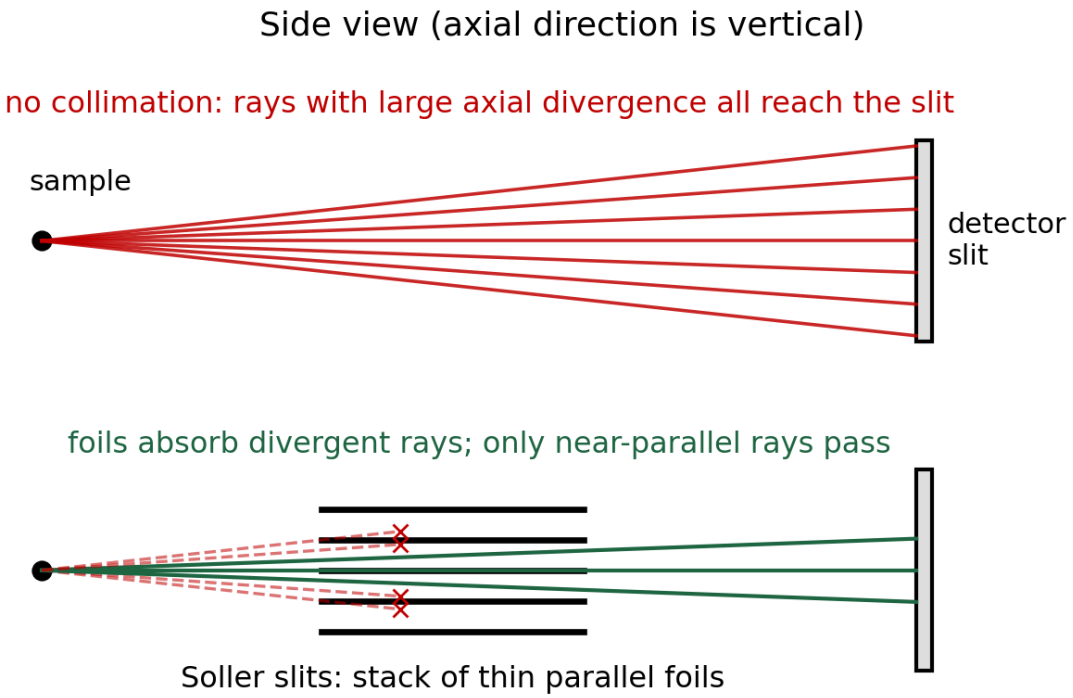
Result: every low-angle peak leans left



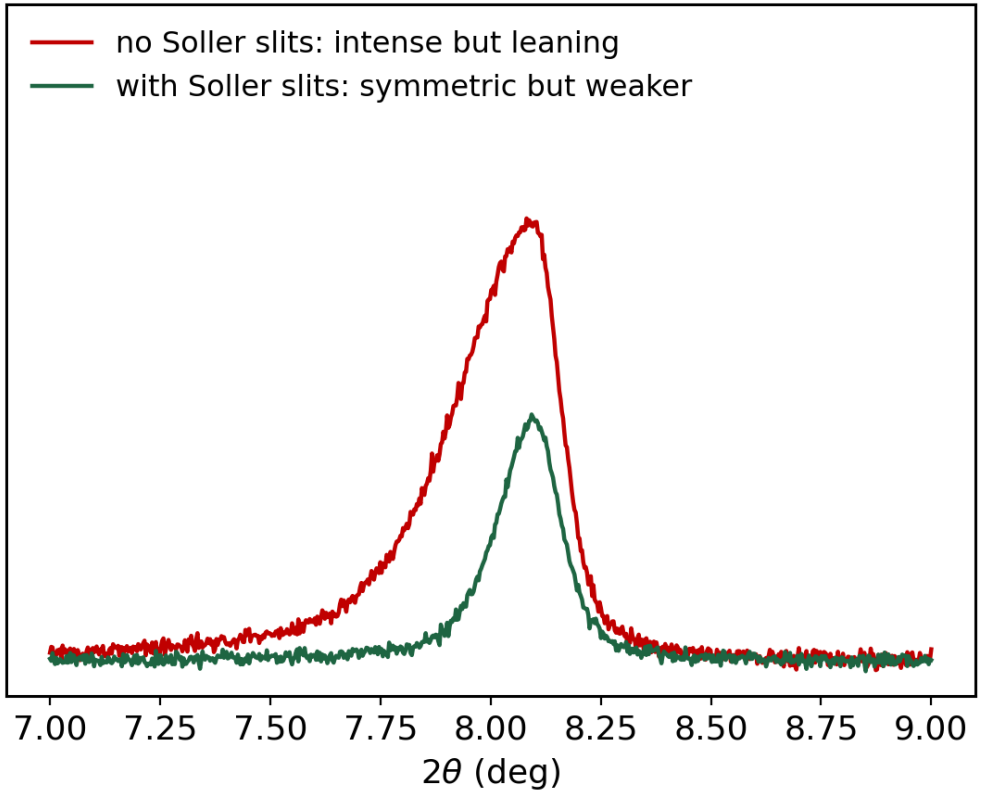
- Diffraction makes cones; on the detector they are rings, and low-angle rings are small and strongly curved
- The receiving slit is straight and tall: scanning outward, its tips clip the ring before its center gets there, so counts arrive early and pile up on the low-angle side
- The early shift scales as $H^2/2r$, which is why the lean is worst at low 2θ (and on high-resolution instruments)
- Model it with the Finger-Cox-Jephcoat correction (built into GSAS-II); never absorb it into peak width

Soller slits tame axial divergence, at the cost of intensity

Side view (axial direction is vertical)

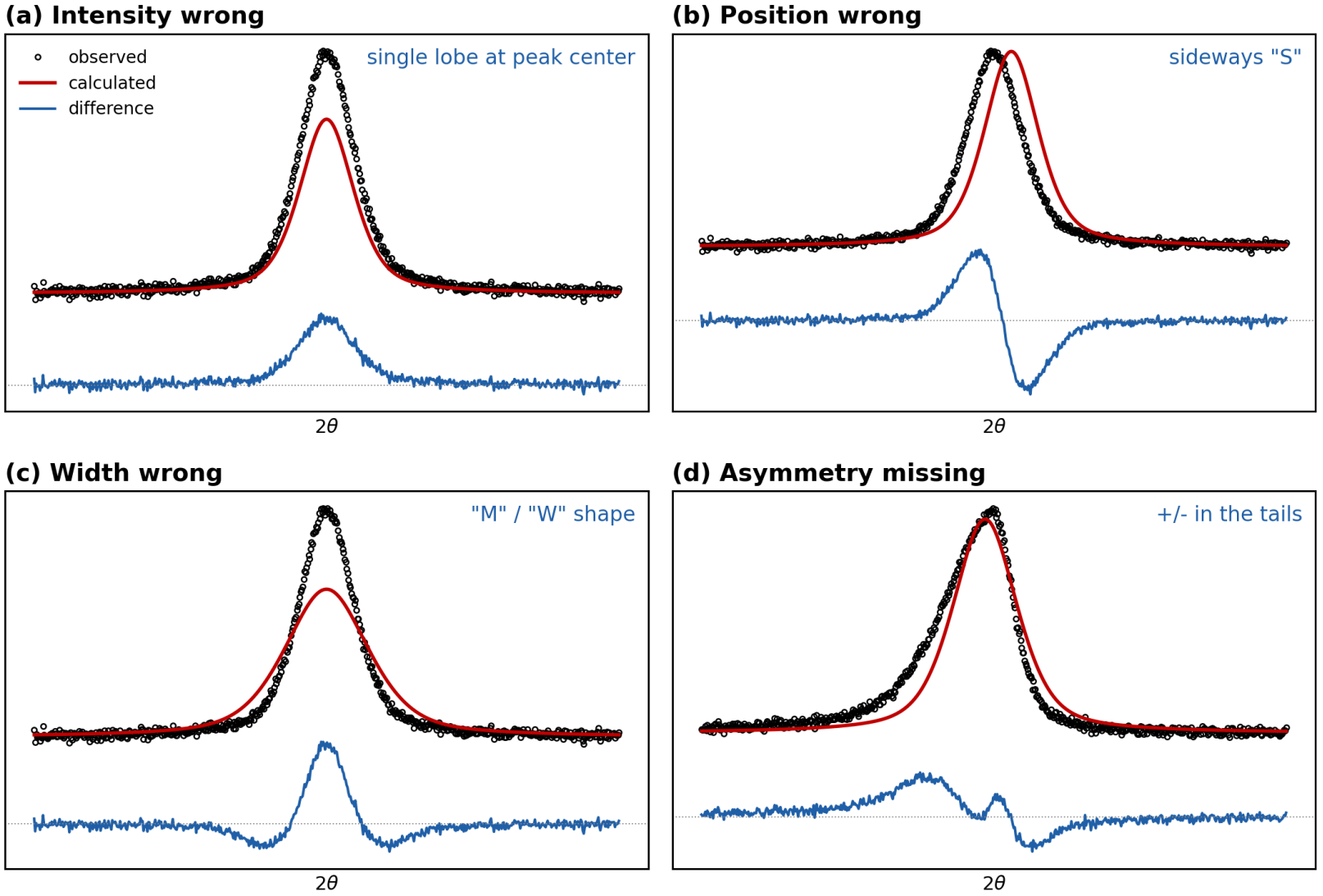


The trade: cleaner shape, fewer counts



- A stack of thin parallel foils limits out-of-plane divergence to roughly (foil spacing) / (foil length)
- Less axial divergence = shorter low-angle tail and more accurate peak positions
- The price is counts: every blocked ray is simply lost, so counting times grow
- Some asymmetry always survives, so the Finger-Cox-Jephcoat correction still applies

The difference curve tells you which parameter is wrong



Real misfits are superpositions of these four fingerprints (after McCusker et al., J. Appl. Cryst. 32, 36, 1999)

Rp, Rwp, and chi-squared put numbers on the fit

- Rp: plain difference between observed and calculated intensities
- Rwp: weighted, so strong peaks matter more than weak ones; this is what least squares minimizes
- **Common guideline: Rwp < 10% or $\chi^2 < 4$ for a good fit**

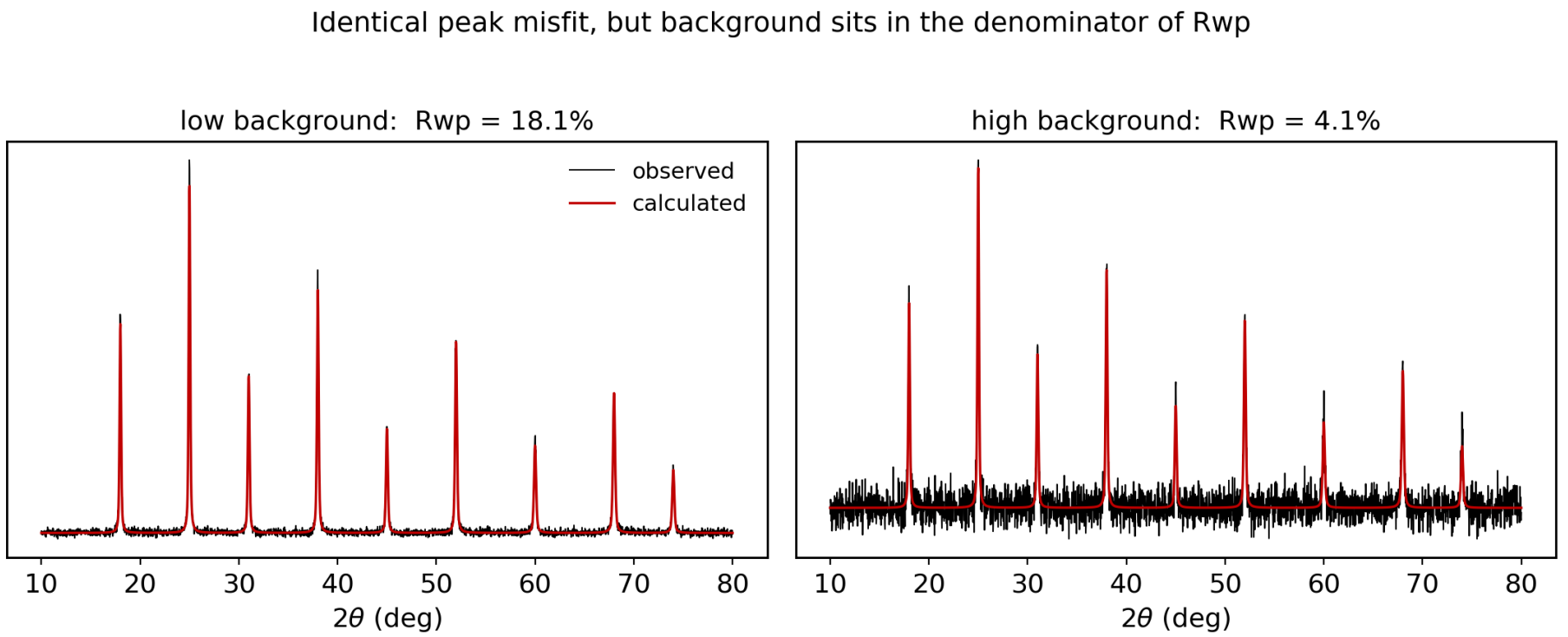
$$R_P = \frac{\sum |y_{io} - y_{ic}|}{\sum y_{io}} \qquad R_{wp} = \left[\frac{\sum w_i (y_{io} - y_{ic})^2}{\sum w_i y_{io}^2} \right]^{1/2}$$

$$GOF = \chi^2 = \left[\frac{R_{wp}}{R_{\exp}} \right]^2$$

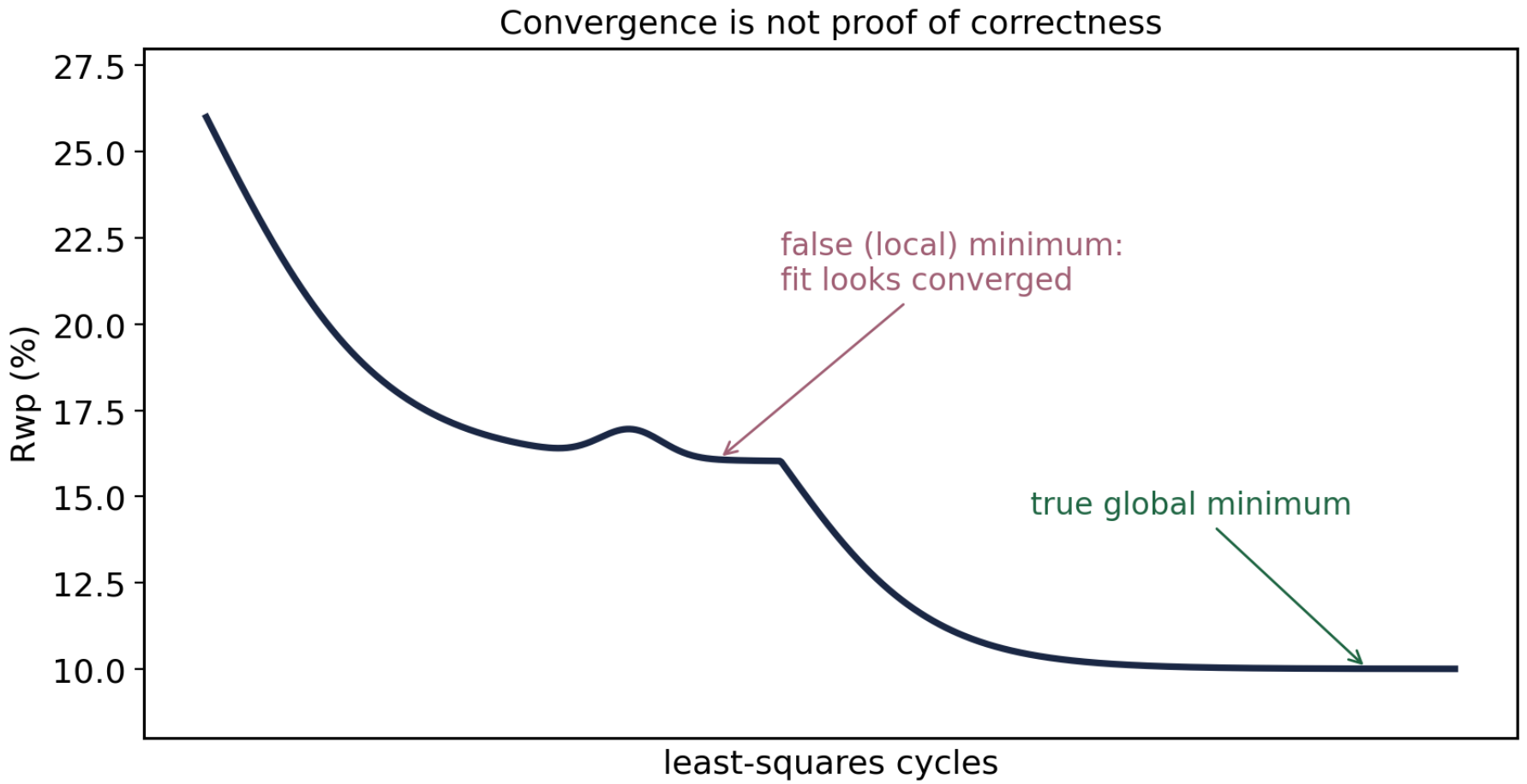
prism.mit.edu/xray/RietveldGOF.pdf

A good Rwp can hide a bad fit, and vice versa

- Background sits in the denominator: high background flatters Rwp for the very same peak misfit
- Never compare Rwp across instruments: TOF neutron at 3% and lab XRD at 10% can be equally good fits
- Your floor is the Le Bail Rwp; judge fits with the plot and chemical sense, not the number (Toby, 2006)



Convergence to a minimum is not proof you found THE minimum

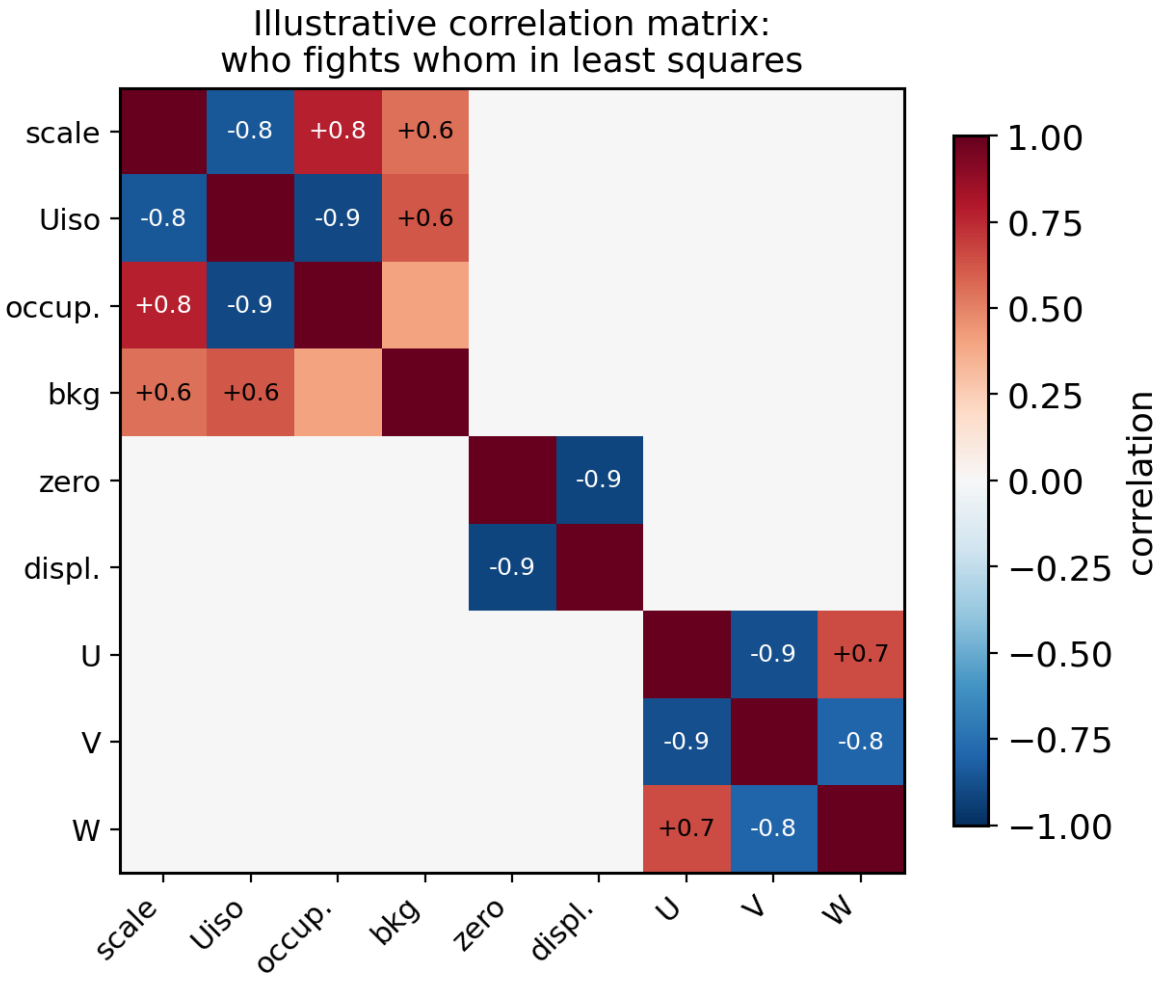


"A Rietveld refinement is never finished, only abandoned." P. W. Stephens

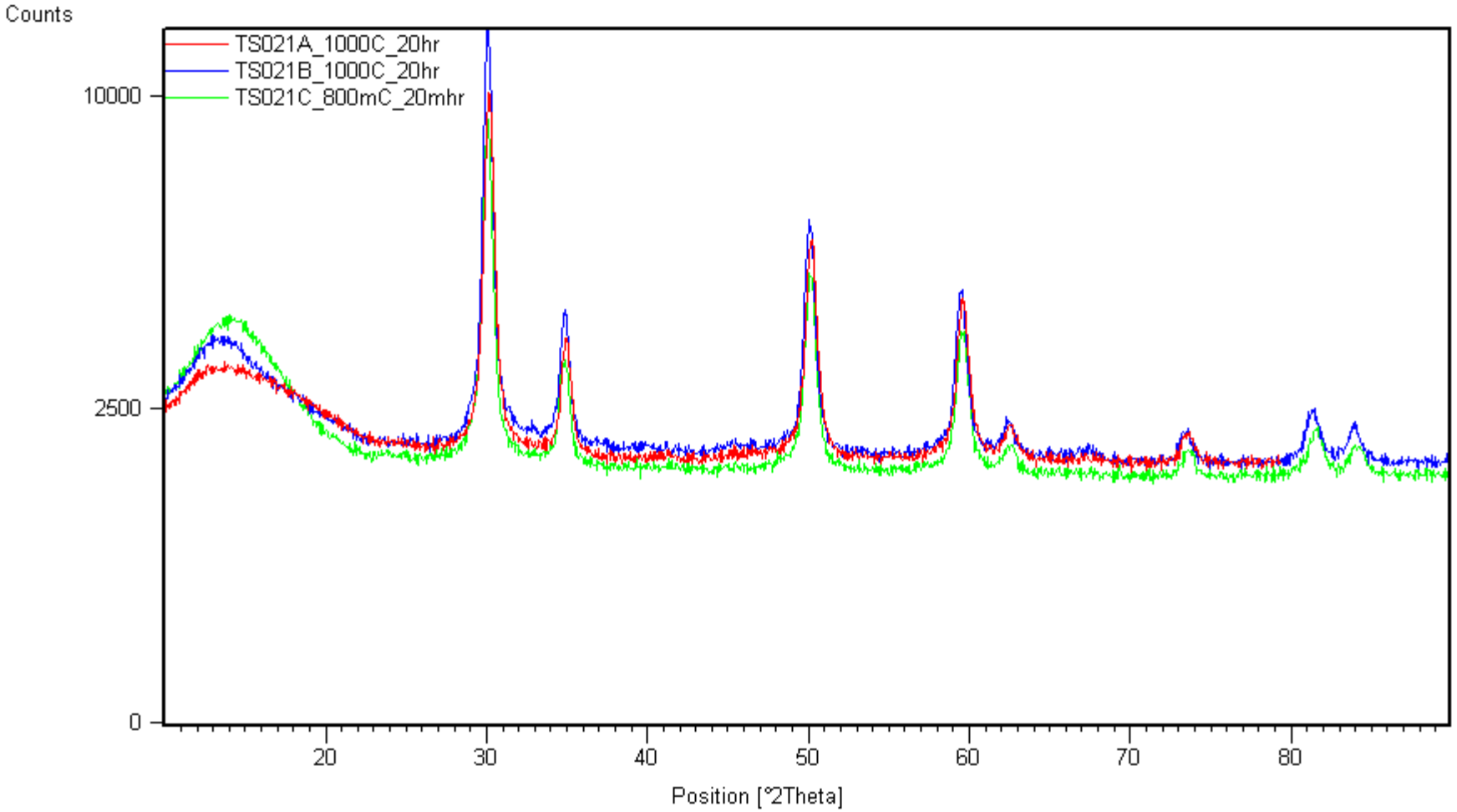
Refinement turn-on order is correlation management

1. Scale + background
2. Lattice + zero (or displacement)
3. Profile: W first, then U, V, X, Y
4. Positions: heavy atoms, then light
5. Uiso (constrain similar atoms early)
6. Occupancies last, then all together

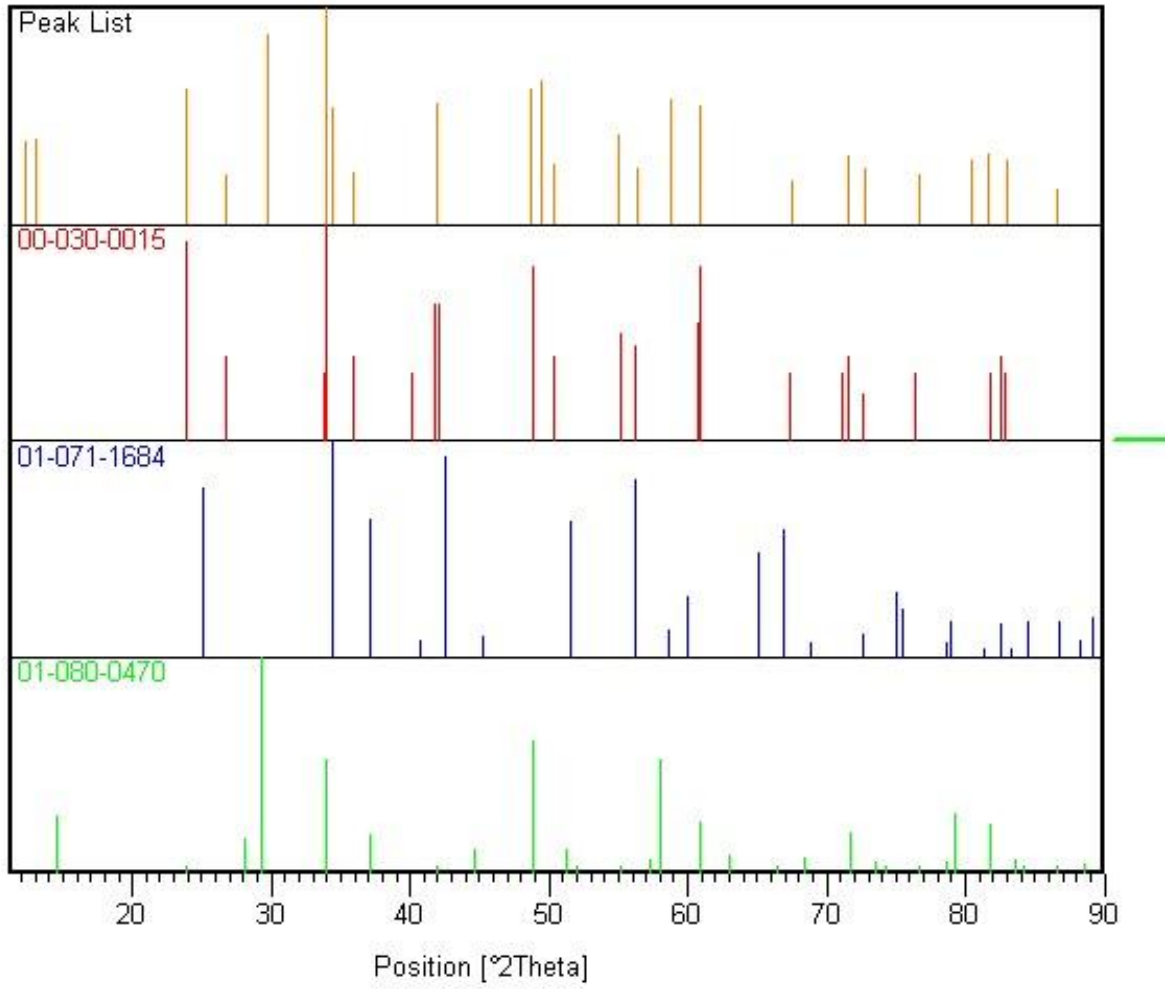
Rule of thumb: observations per parameter ≥ 3, ideally 5



Multiphase analysis starts by matching the peak list to phases



A three-phase Rietveld fit resolves the mixture directly

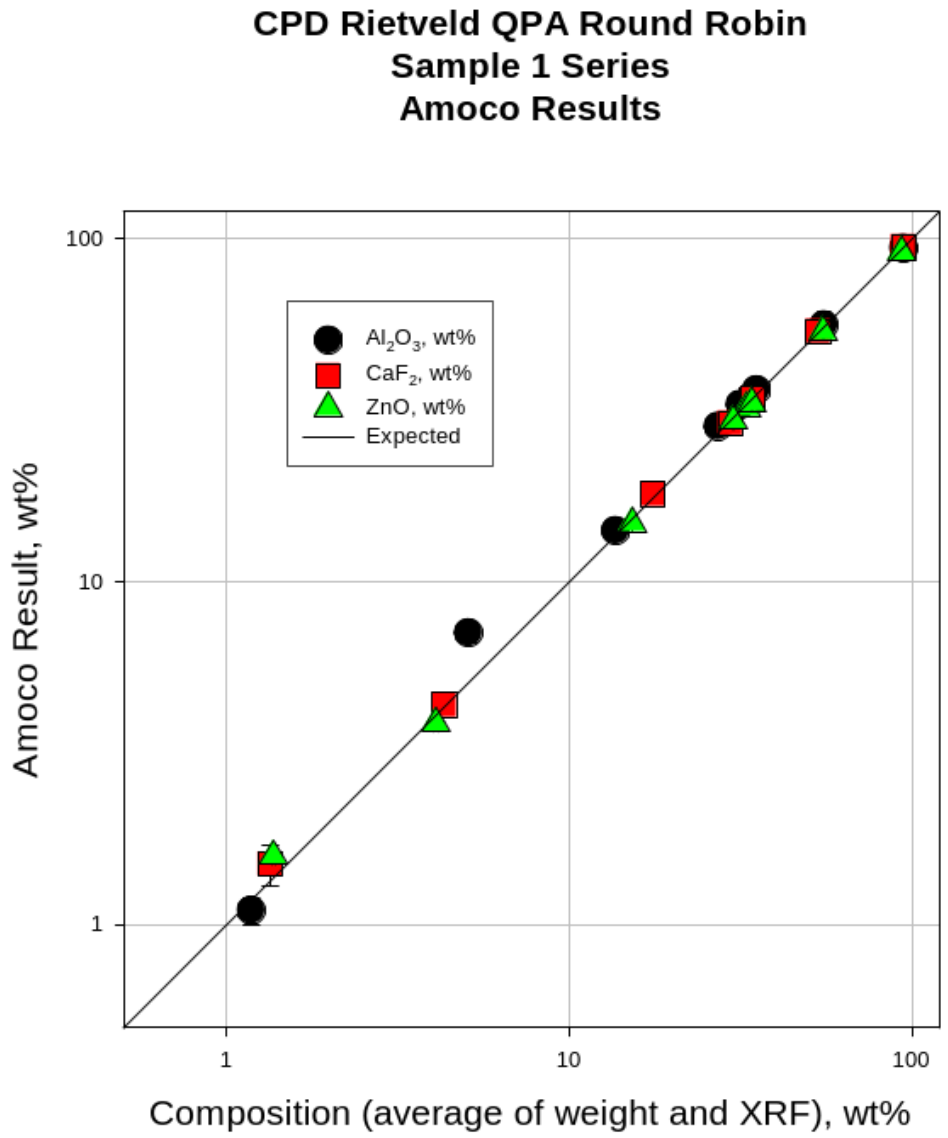


The SMZ relation converts scale factors into weight fractions

- S is the phase scale factor, proportional to the number of unit cells measured
- M is the molecular weight
- Z is the number of formula units per unit cell
- The product SMZ is proportional to the concentration of each phase

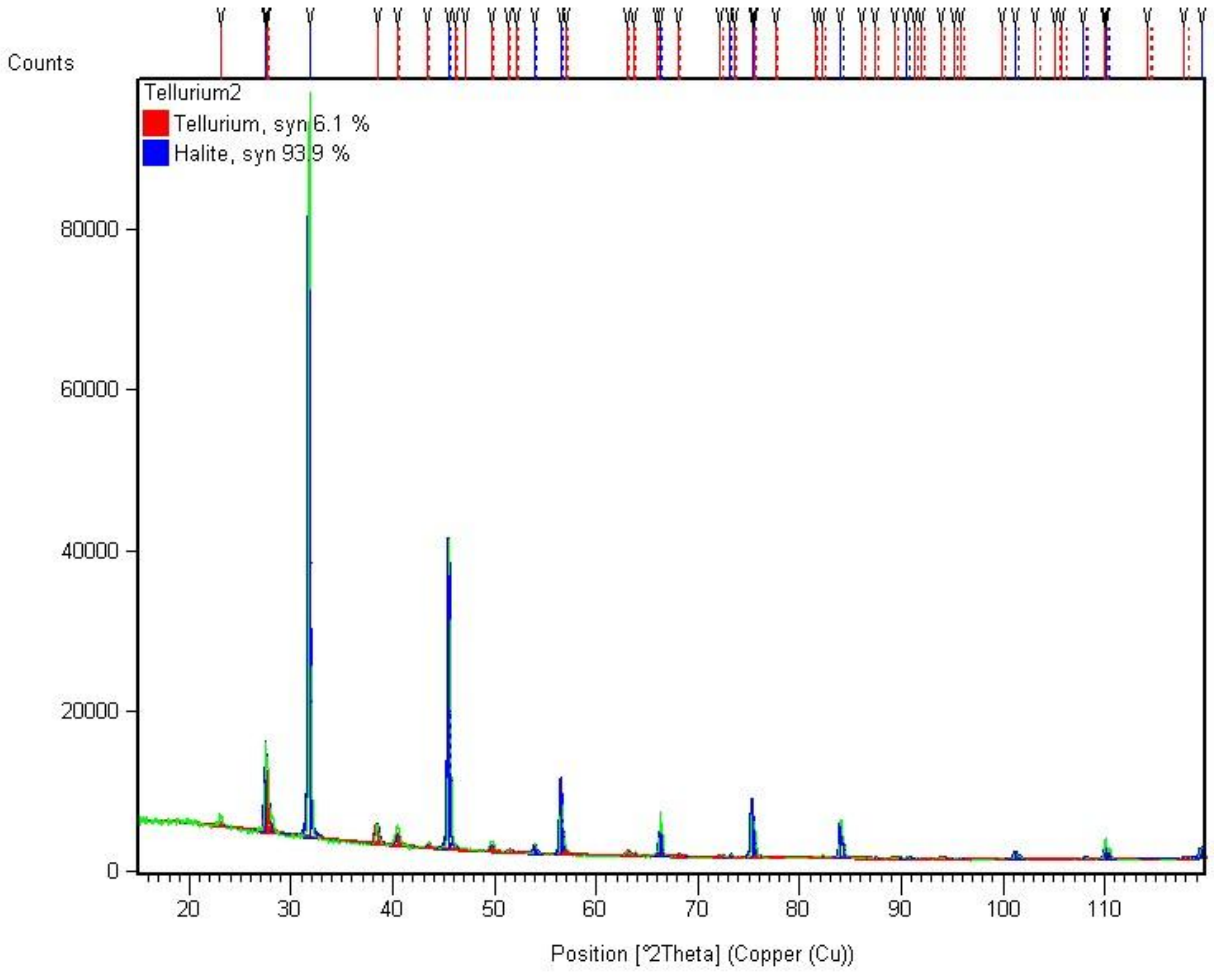
$$X_\alpha = \frac{S_\alpha M_\alpha Z_\alpha}{\sum_i S_i M_i Z_i}$$

Round-robin studies show quantitative analysis is accurate to a few percent



James Kaduk, ICDD, Rietveld Refinement Workshop, Oct 2010

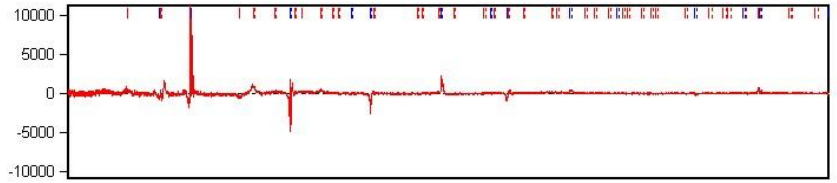
Rietveld quantifies 6% tellurium recovered from a seawater medium



93.9% NaCl

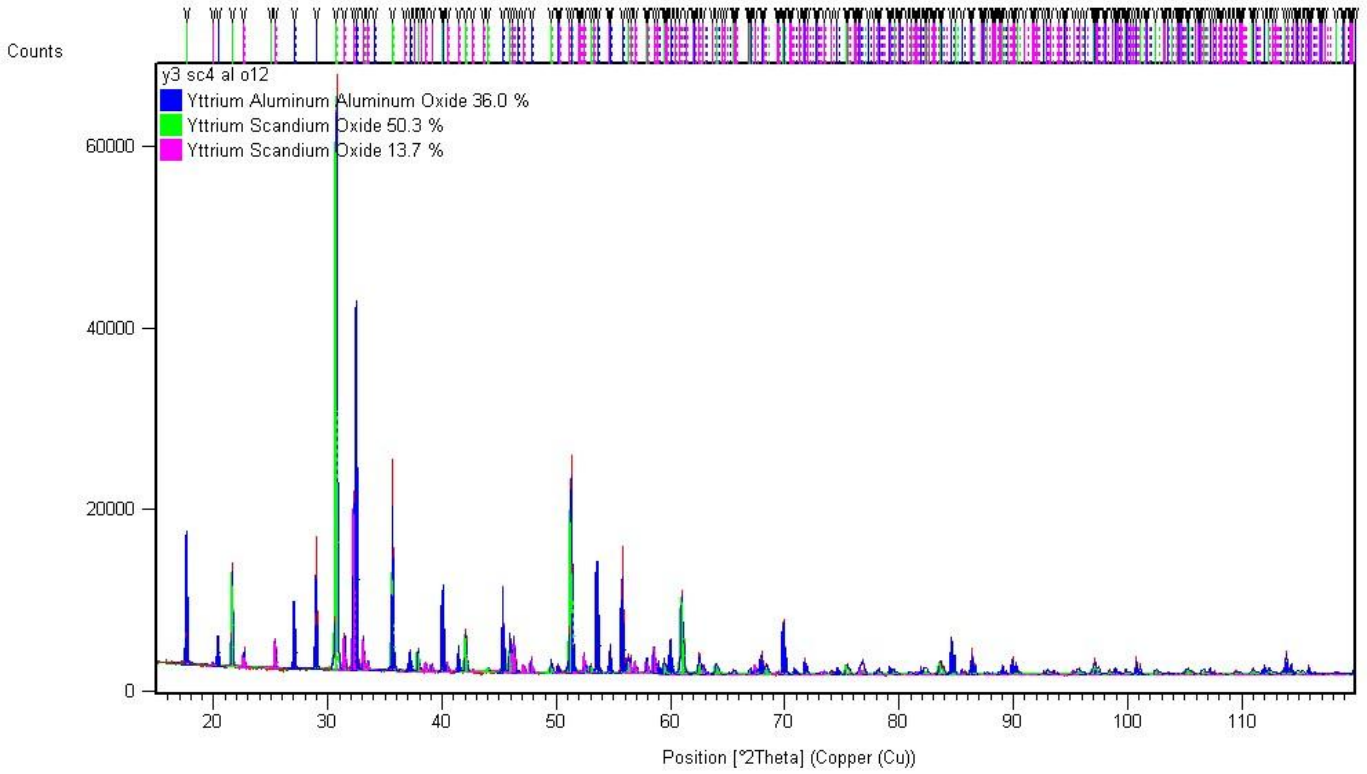
6.1% Te

Rwp = 6.9%



Error sources: absorption (transparency), sample volume (more powder)

The intended Y3Sc4AlO12 actually crystallized as three phases

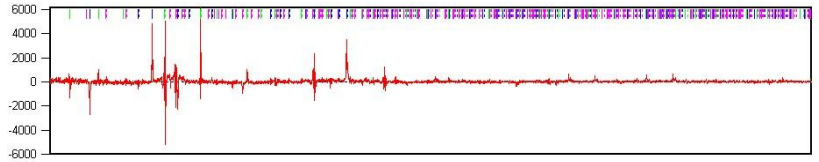


36% YAG (Ia-3d)

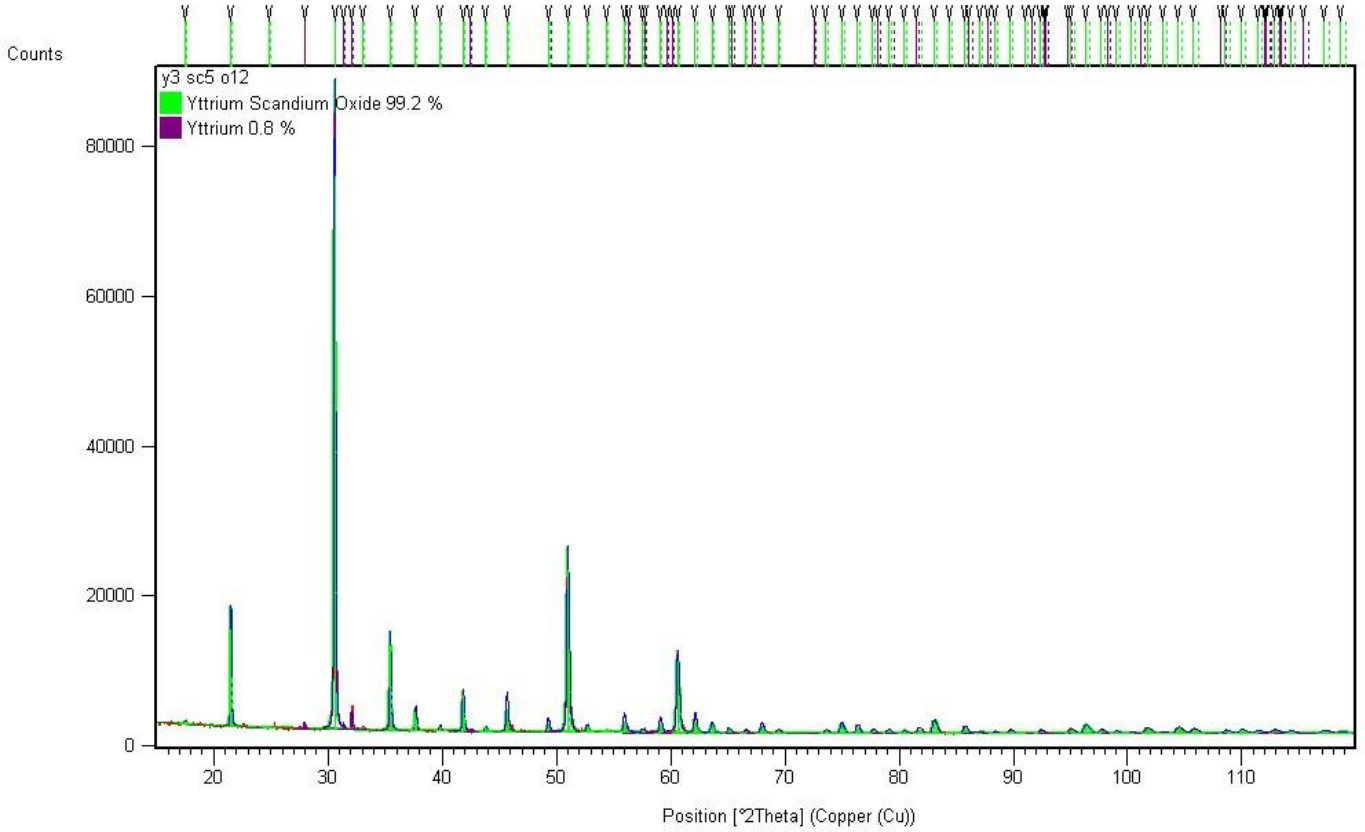
50.3% YScO3 (Ia-3)

13.7% YScO3 (Pnma)

Rwp = 5.3%



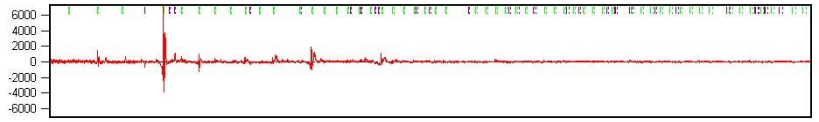
Y3Sc5O12 instead formed nearly phase-pure bixbyite



99.2% Y3/4Sc5/4O3 (Ia-3)

0.8% Y (P63/mmc)

Rwp = 4.7%



The Debye equation extends analysis to amorphous content

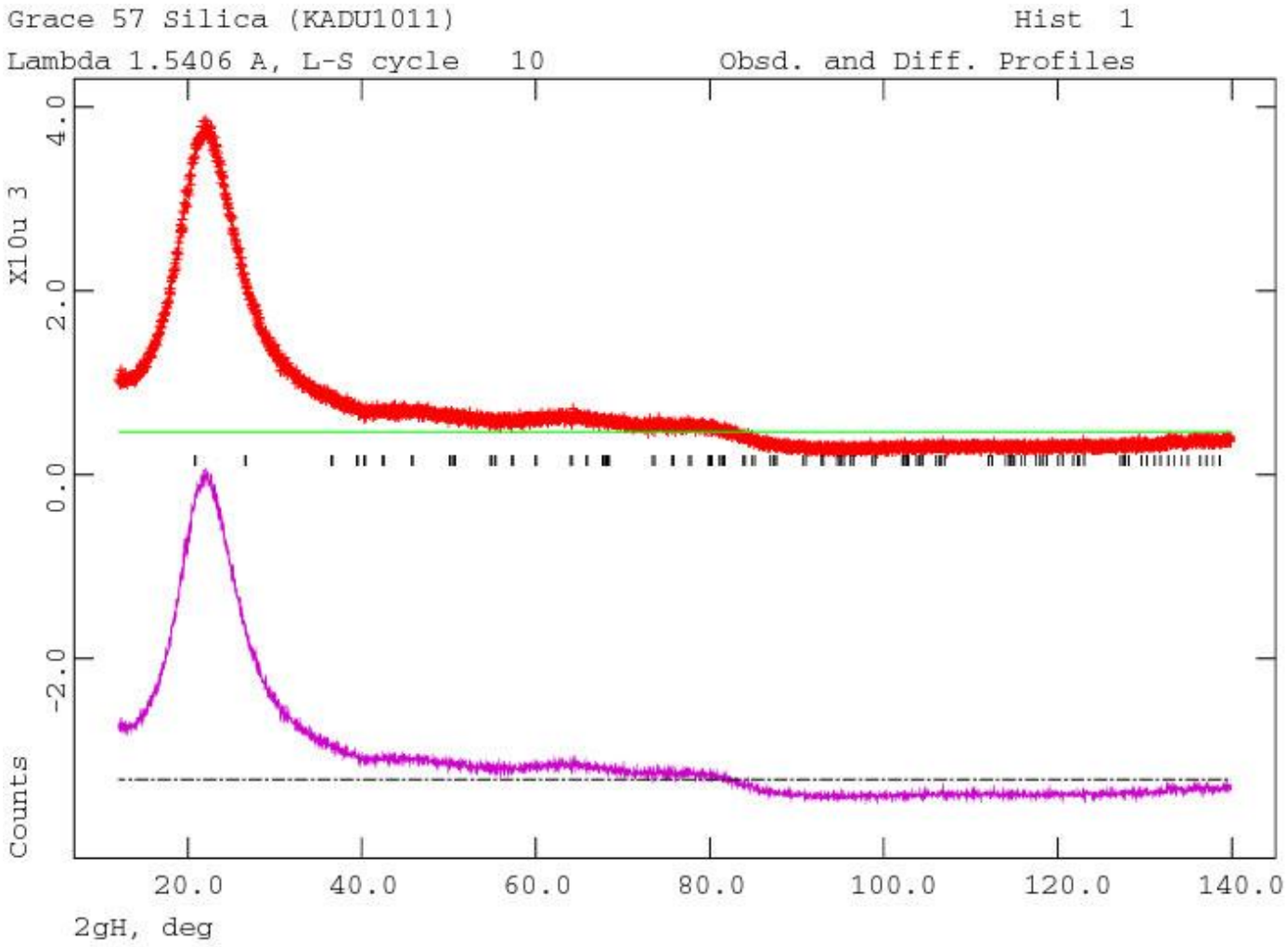
- Percent amorphous material can be determined if a standard is added in a known amount
- Terms describe interatomic distance and vibration amplitude
- A radial distribution function is needed to determine bond distances for analysis

$$I(\theta) = \sum_i f_i^2(\theta) + 2\sum_i \sum_{j>i} f_i(\theta) f_j(\theta) \frac{\sin\left(4\pi r_{ij} \sin\theta/\lambda\right)}{4\pi r_{ij} \sin\theta/\lambda}$$

$$A \frac{\sin(QR_i)}{QR_i} \exp\left(-\tfrac{1}{2} B_i Q^2\right)$$

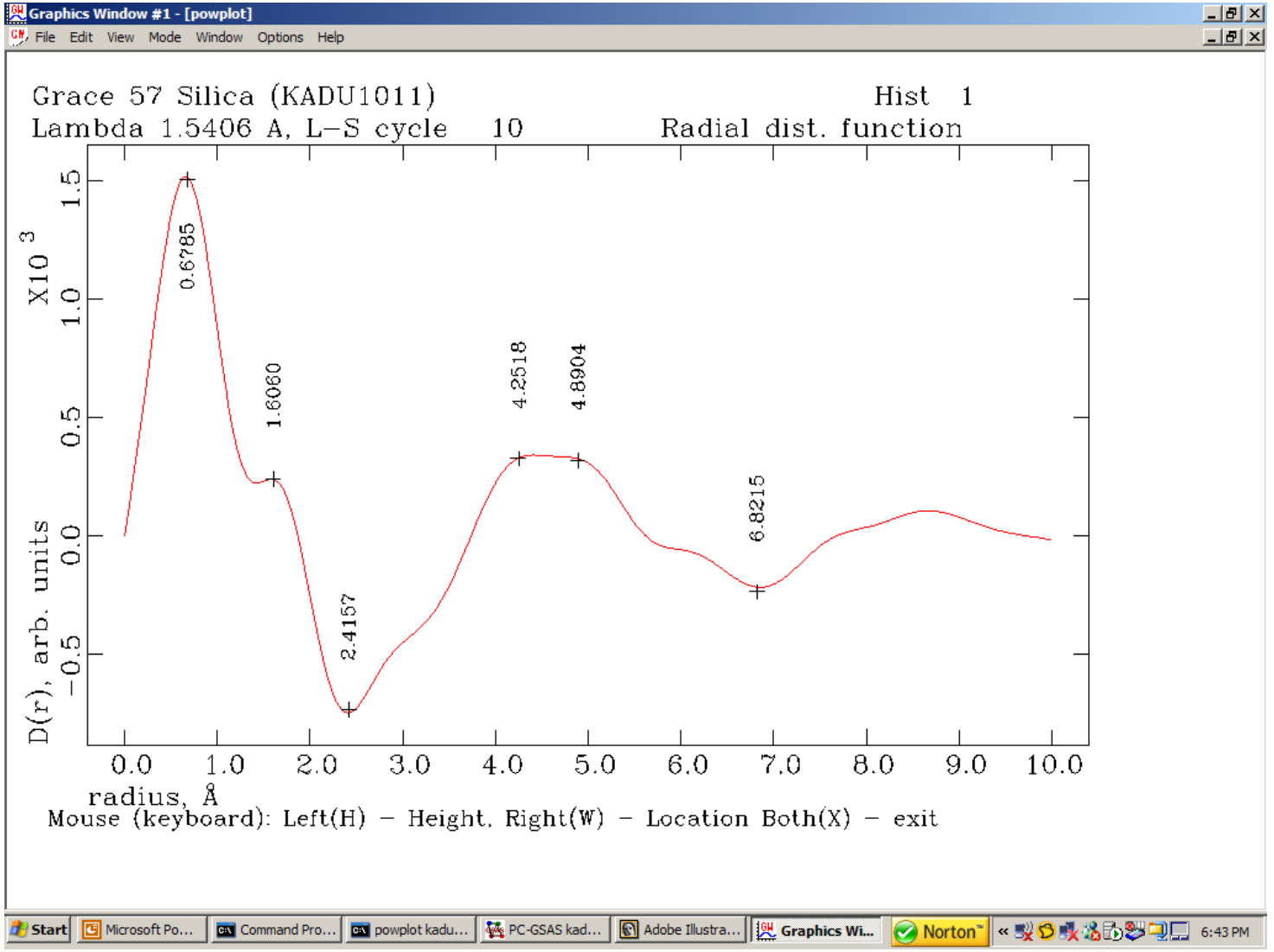
A: amplitude R_i: distance B_i: vibration

Glassy silica's diffuse scattering can be fit, not just subtracted

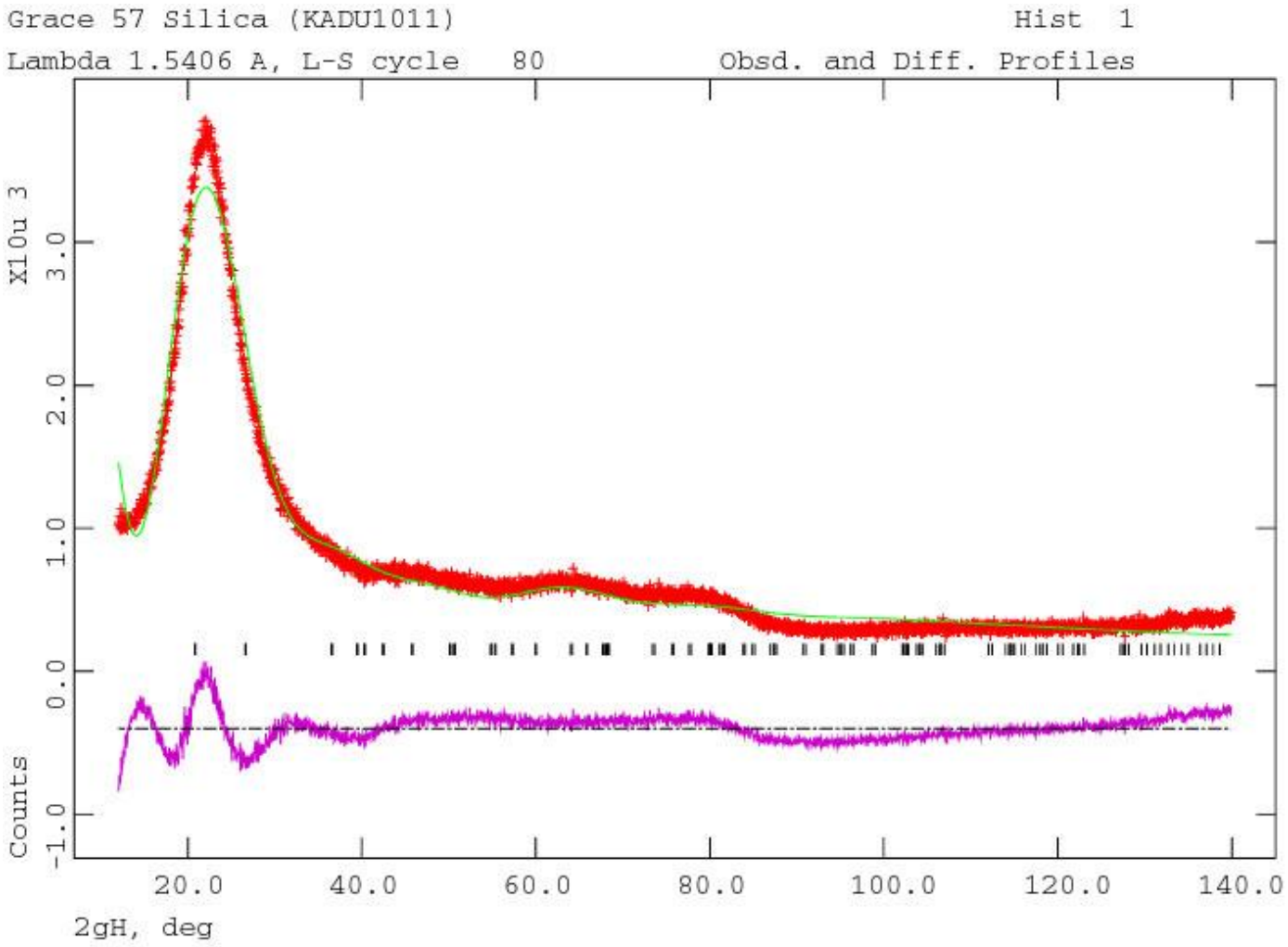


James Kaduk, ICDD, Rietveld Refinement Workshop, Oct 2010

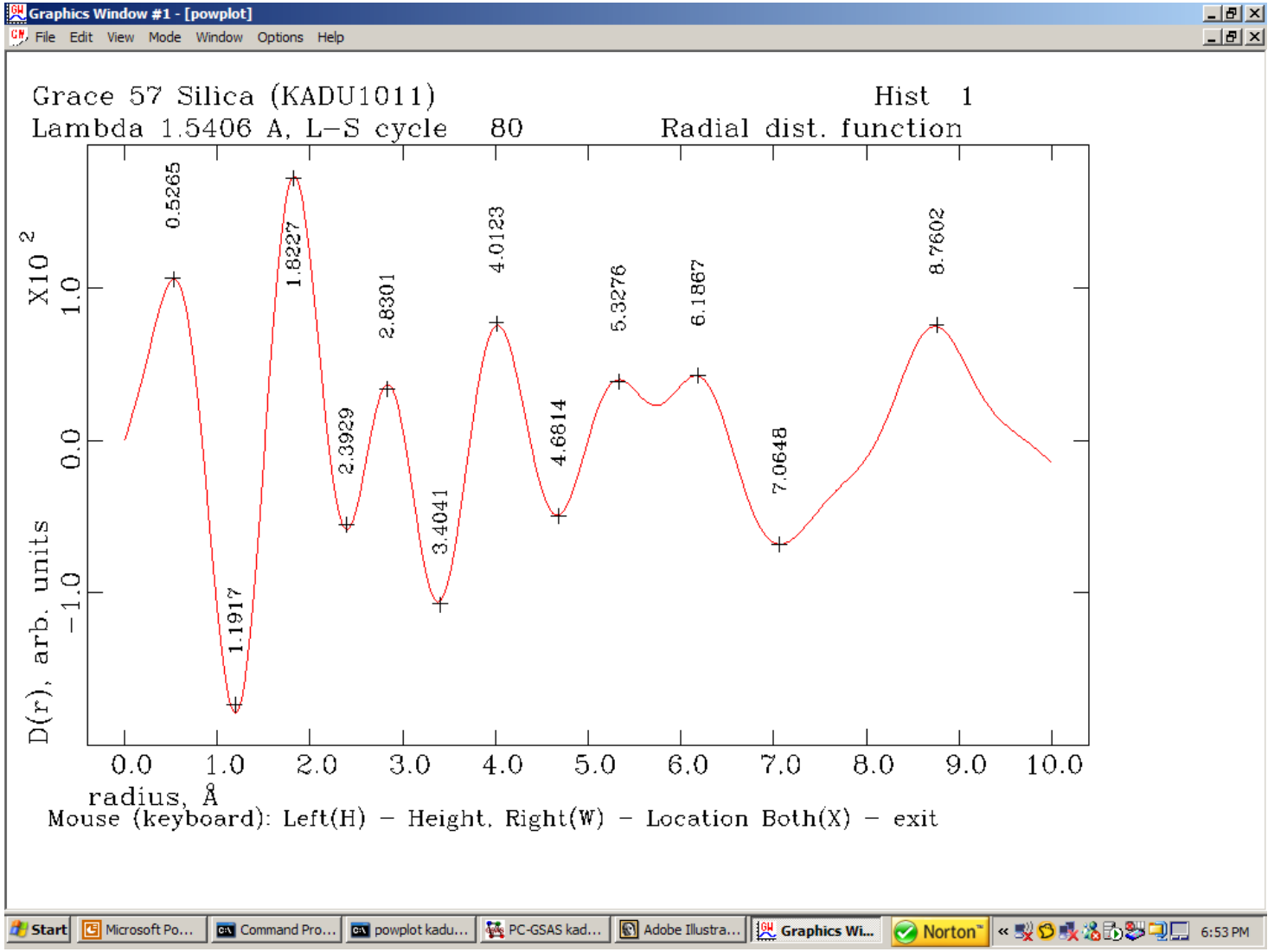
The refined model yields a real-space radial distribution function



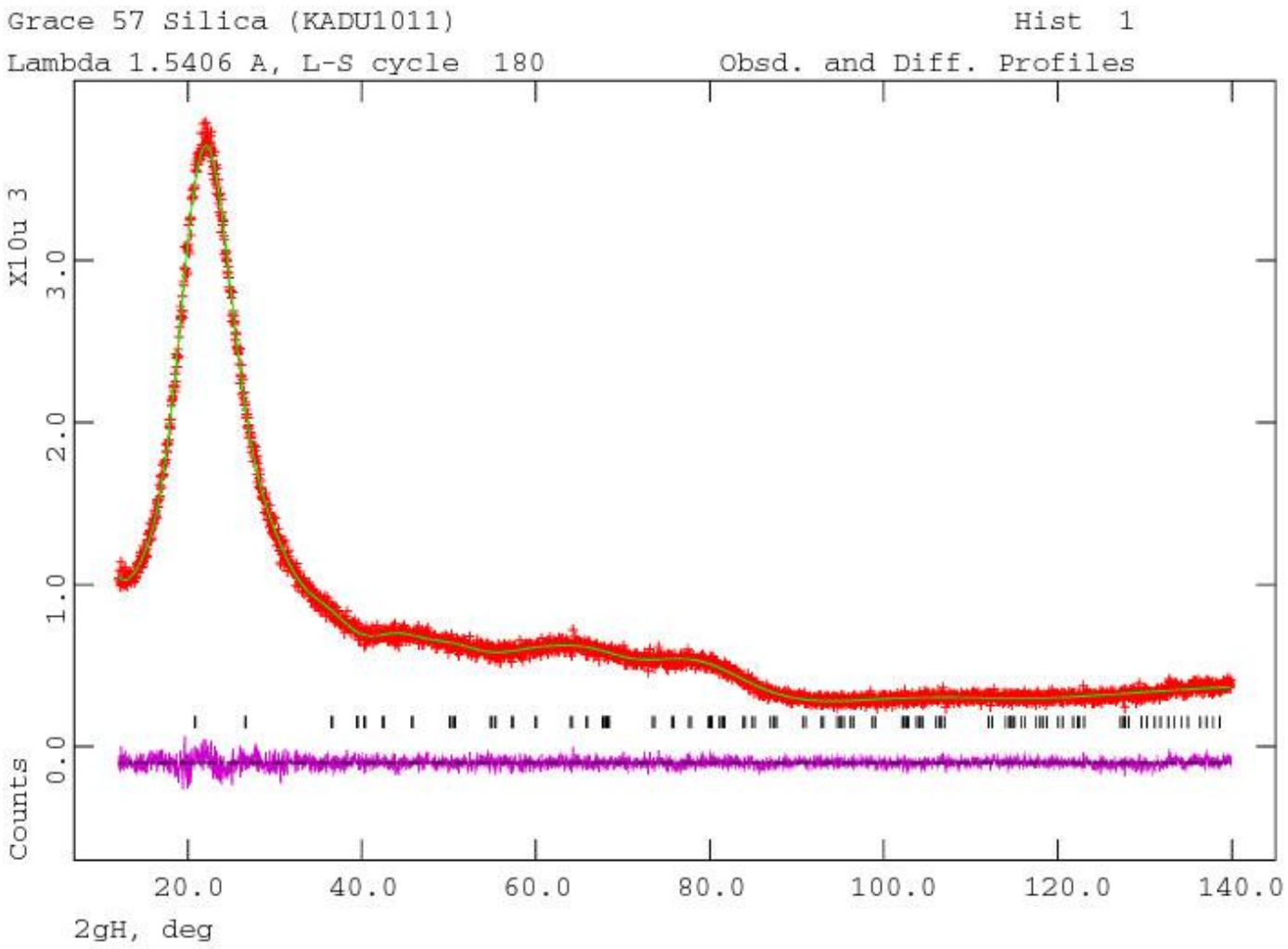
Iterating the Debye model steadily improves the diffuse fit



RDF peaks index directly to coordination shells



The final diffuse fit converges with chi-squared = 1.11



Lab XRD alone cannot settle apatite site occupancies

- Gd8+xCa2+y(SiO4)6O2+3x/2+y system with anion/cation vacancies
- **Lab XRD is insufficient to answer:**
  - Impurity wt%?
  - Volume change?
  - Cation site preference?
  - Meta-prism twist?

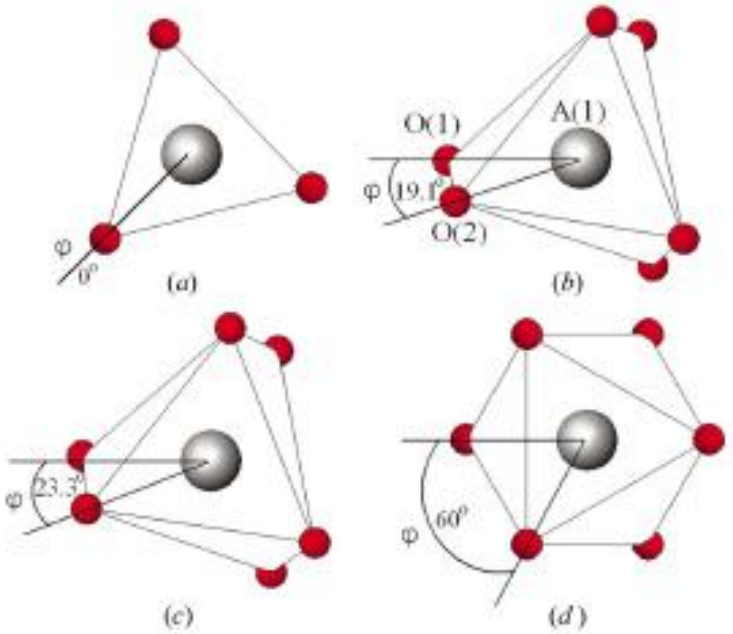
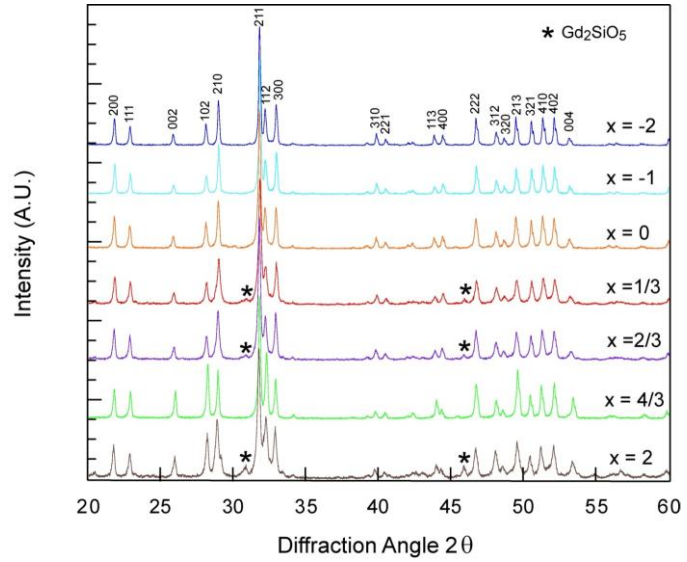
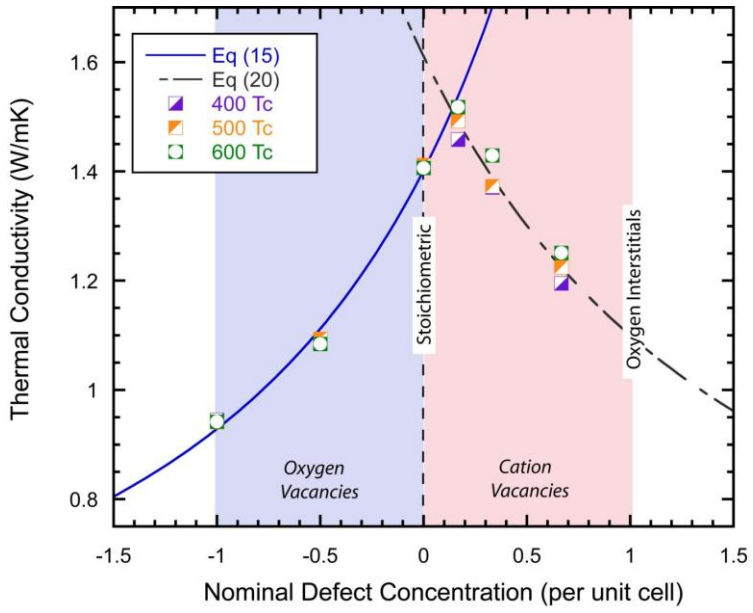


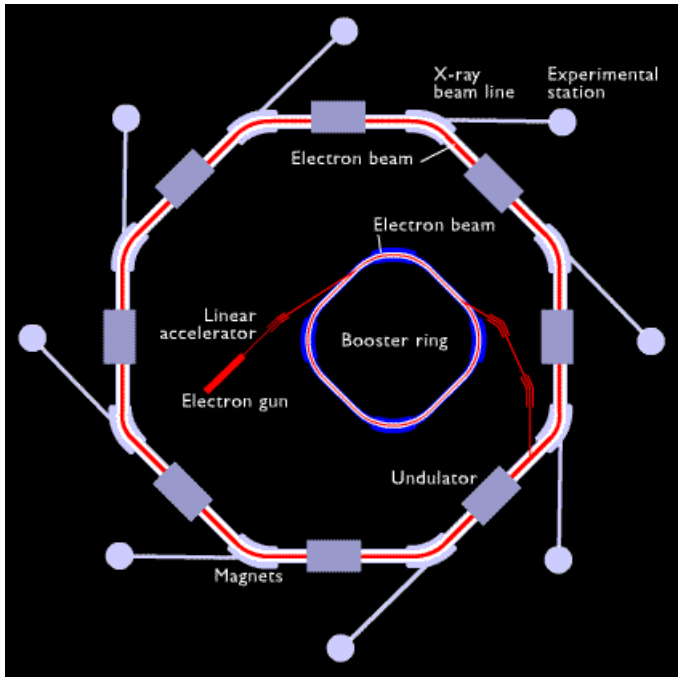
Figure 6
Twist angles of $A(1)O_6$ polyhedra in (*a*) models I and II, (*b*) chlorapatite, (*c*) fluorapatite, and (*d*) model III (as found in glaserite).



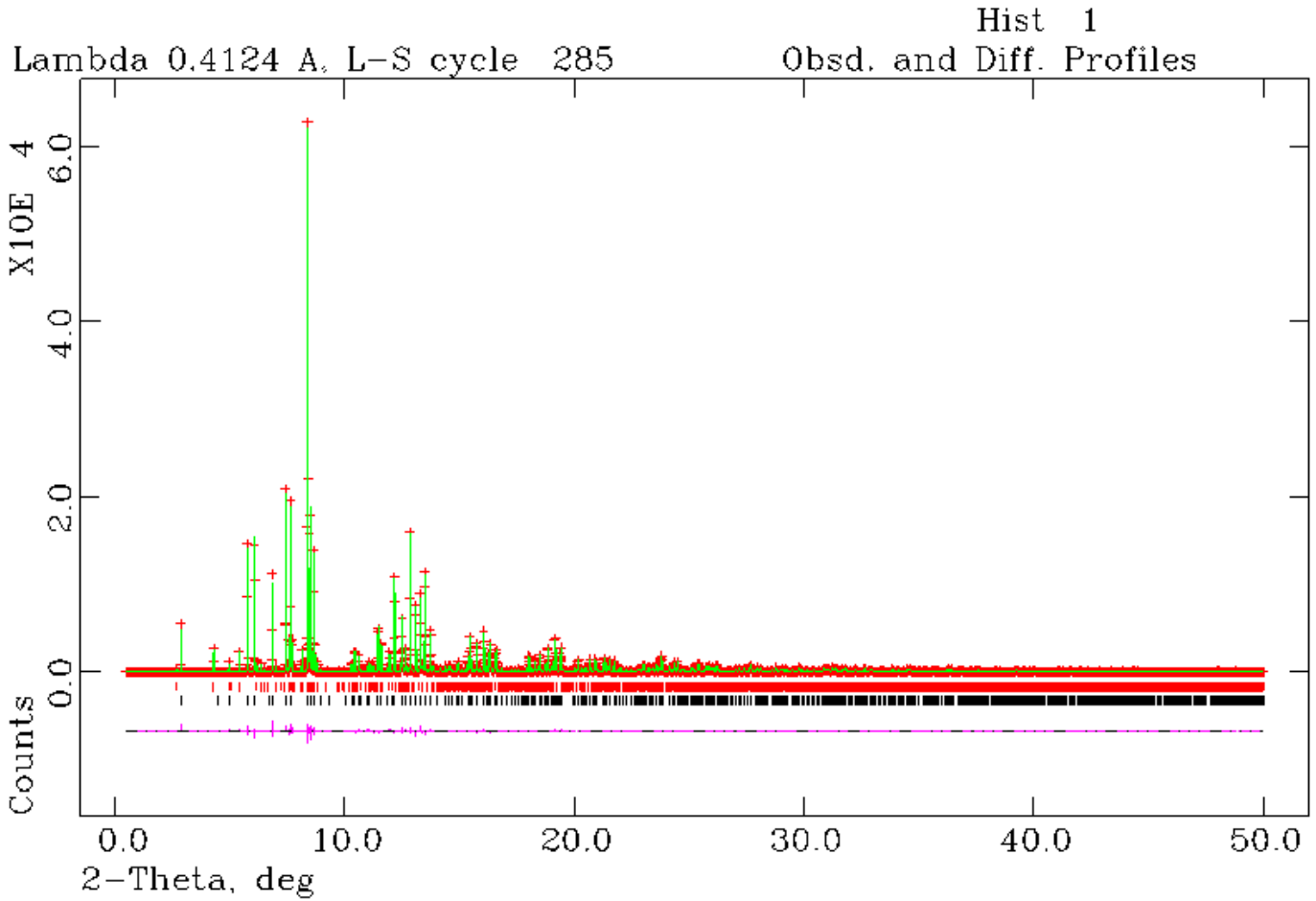
Synchrotron data adds the resolution occupancy refinement needs

- Stoichiometric composition should have the highest thermal conductivity
  - Phase separation?
- **Synchrotron diffraction brings:**
  - Improved signal-to-noise ratio
  - Improved resolution
  - Minimized anomalous scattering





Refinement finds 5% Ca on A1 and a 23.6 degree meta-prism twist



Rwp = 6.29%

85% Gd7.7Ca2.3(SiO4)6O2

14% Gd2SiO5

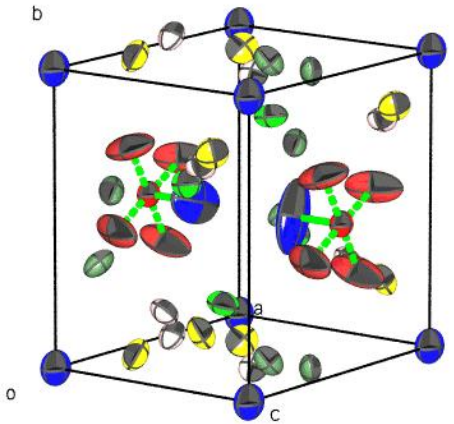
Ca has ~5% on A1

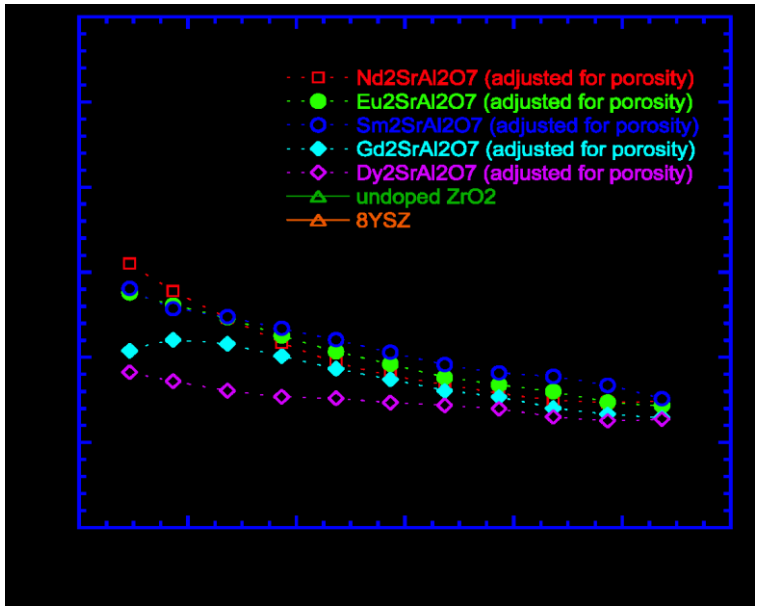
Meta-prism twist 23.6°

Initial composition Gd8.333Ca1.5(SiO4)6O2

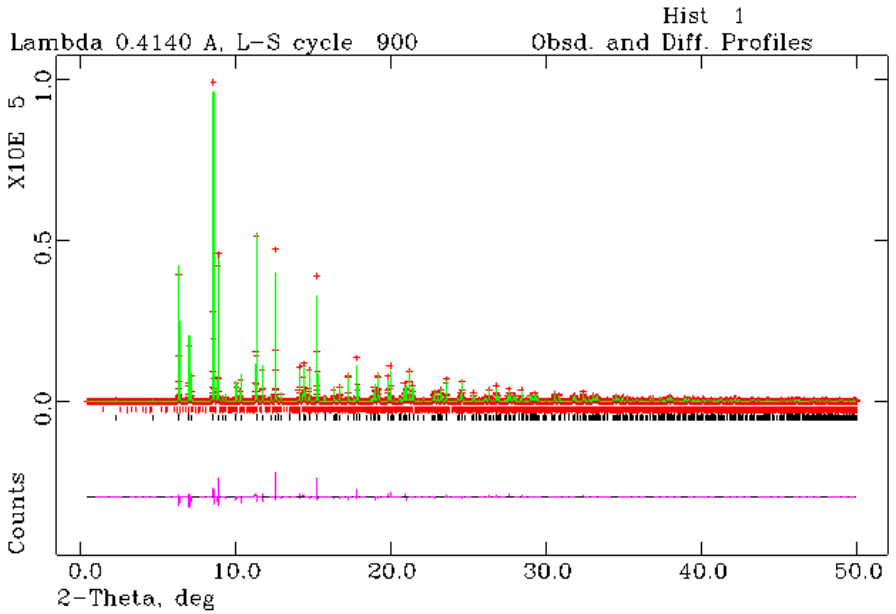
Large refined Uiso values can flag rattling atoms

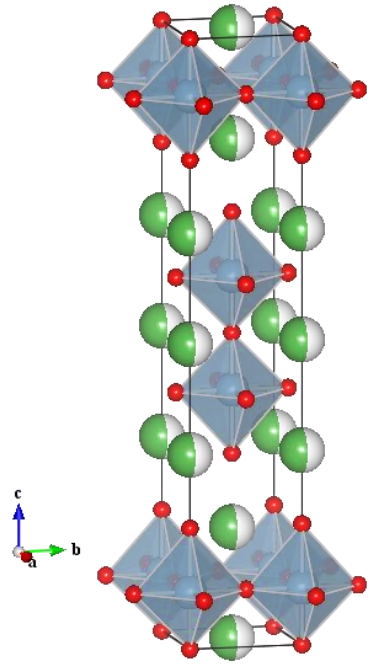
- Thermal vibrations are quantitatively studied in Rietveld refinement
  - Uiso, or Uij for anisotropic vibration
- **Large U values could be due to rattling**
  - An explanation of RE2SrAl2O7 thermal conductivity?





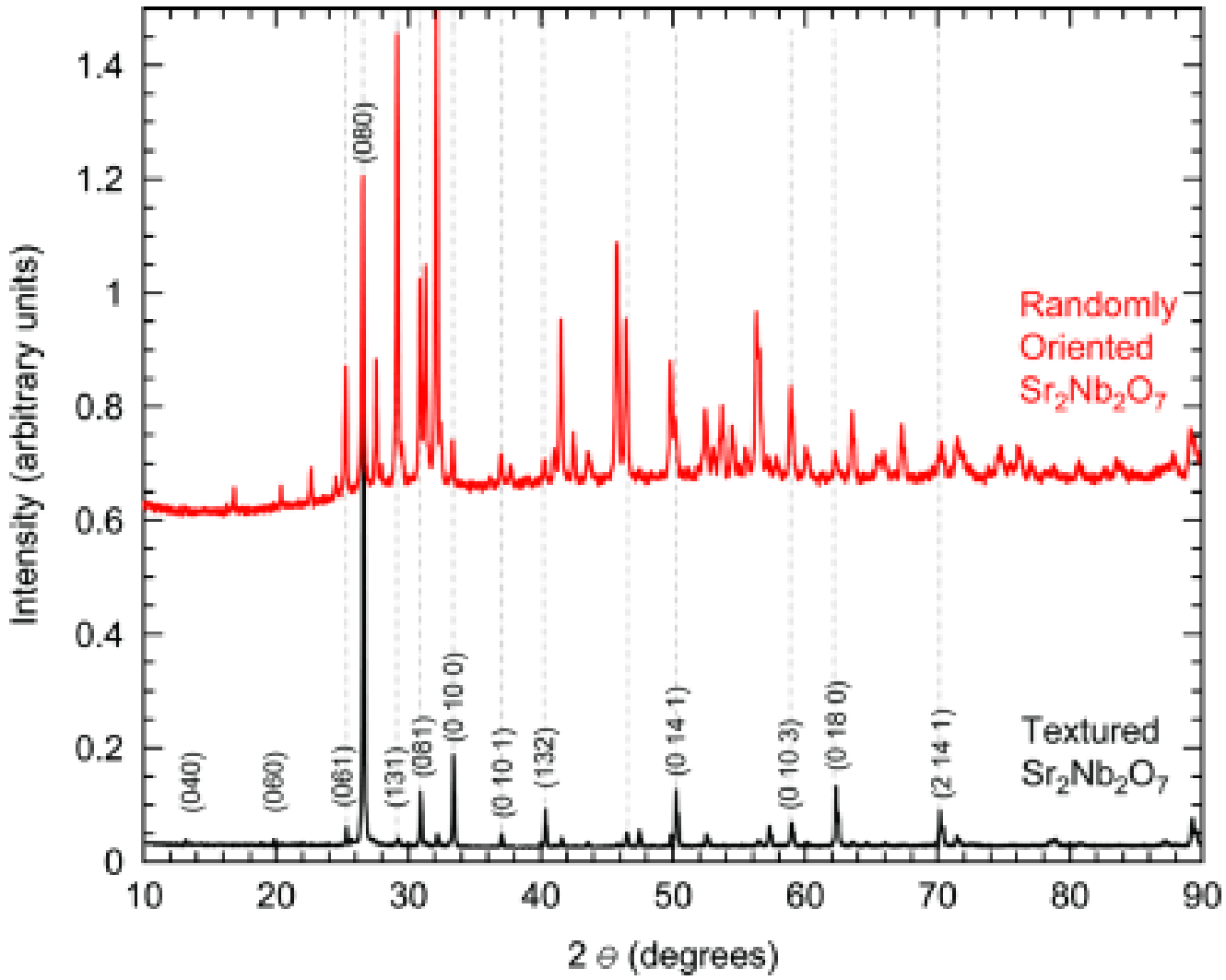
Occupancy and Uiso trends track across the RE2SrAl2O7 series





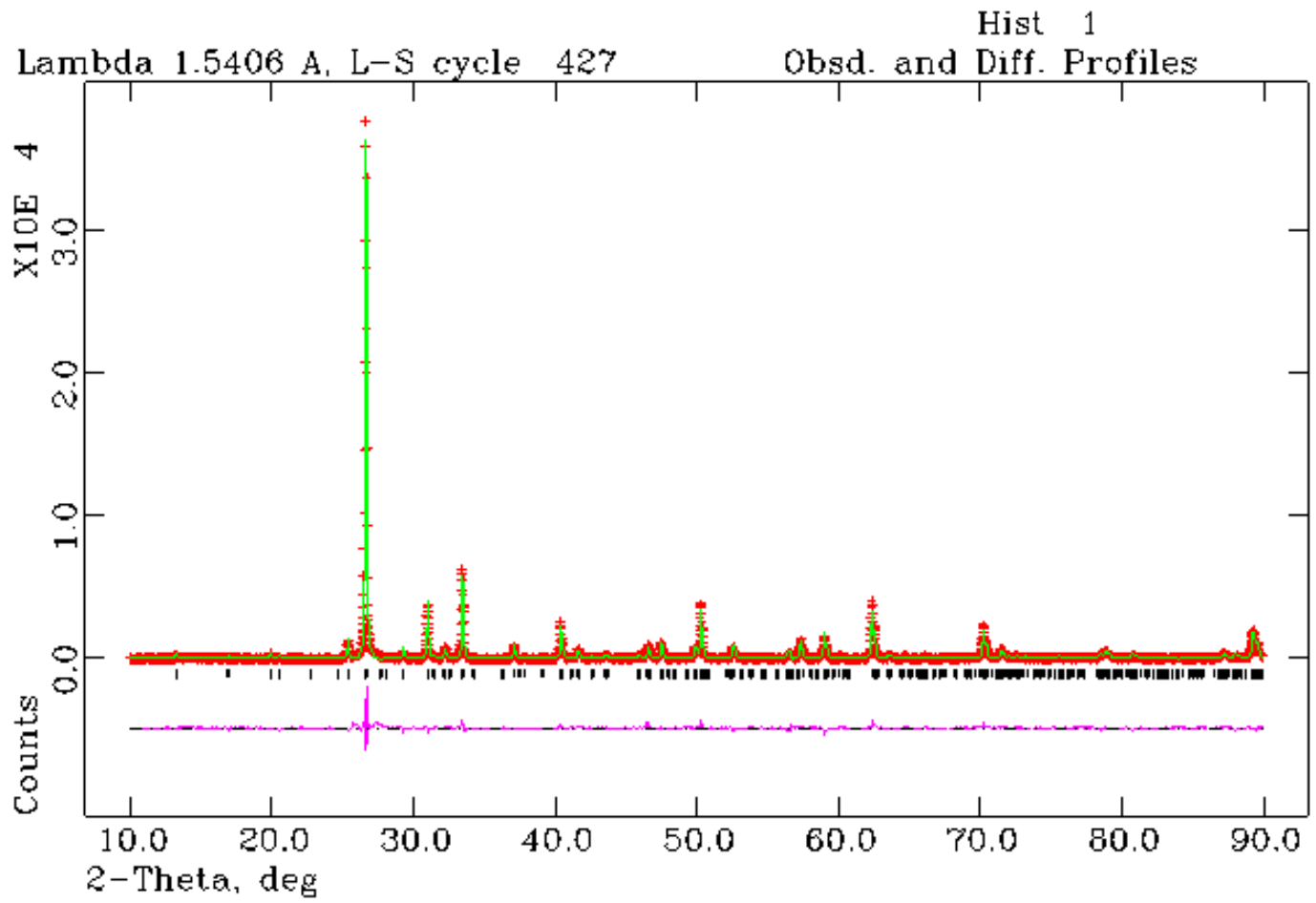
Composition	La2SrAl2O7	Nd2SrAl2O7	Sm2SrAl2O7	Eu2SrAl2O7	Gd2SrAl2O7	Dy2SrAl2O7
Uiso A1	0.11(2)	0.21(5)	0.40(3)	0.20(1)	0.37(3)	0.96(2)
A1 (P2) occupancy	.73	.54	.43	.32	.25	.21
A2 (RS) occupancy	.27	.46	.56	.67	.72	.79

Texture is obvious by eye; Rietveld puts a number on it



Sr2Nb2O7: clearly textured to the b-axis, but what is the preferred orientation factor?

March-Dollase refinement gives Sr2Nb2O7 a texture index of 5.52



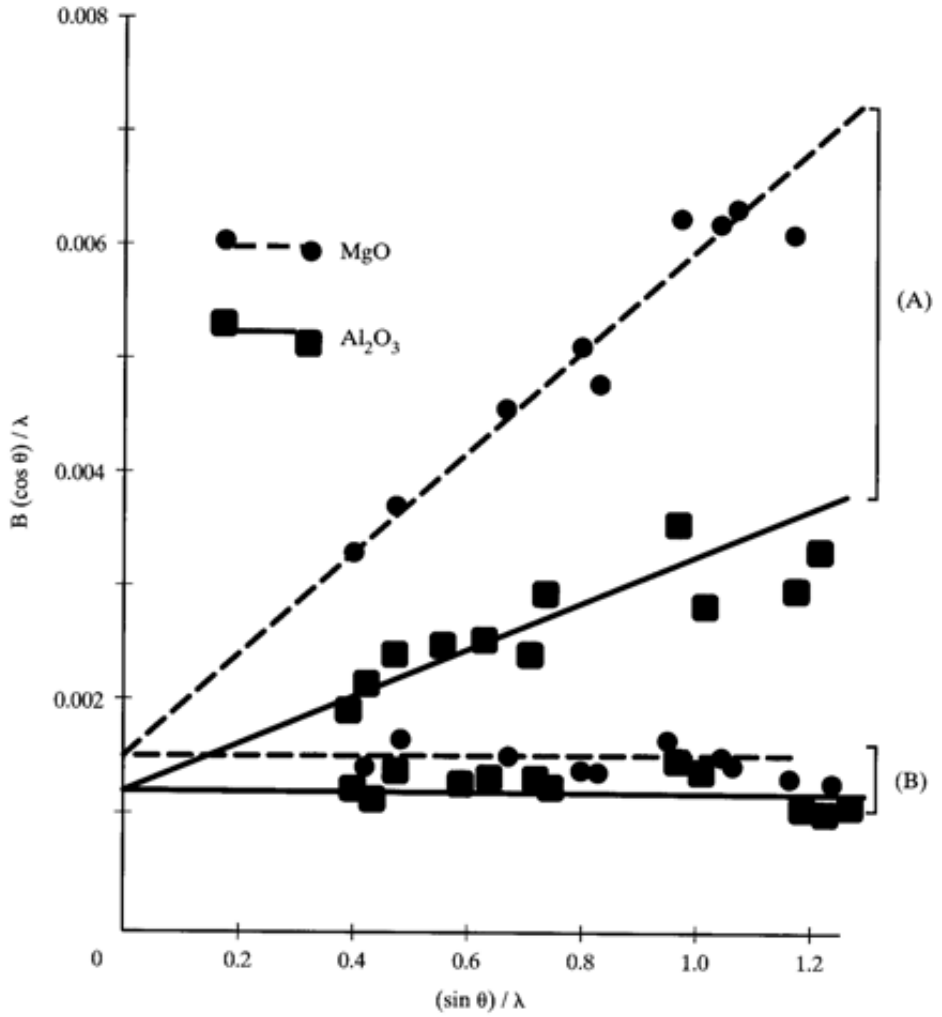
Rwp = 6.9%

Texture index = 5.52

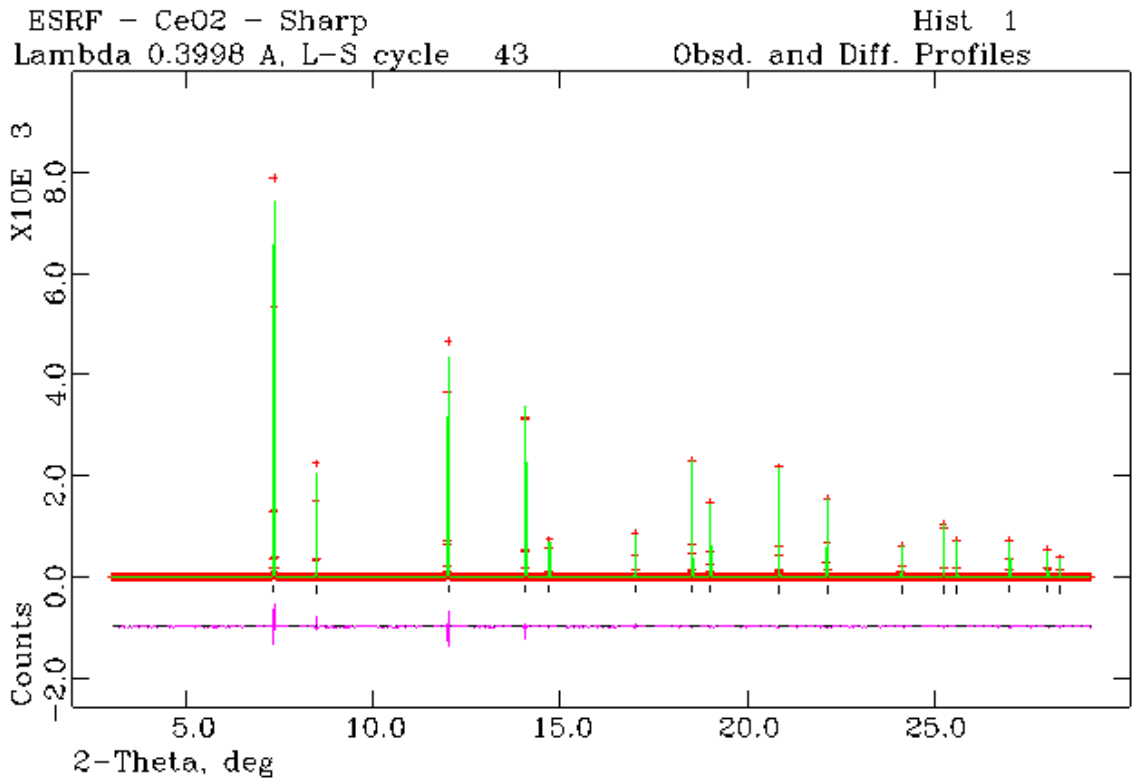
Note: the actual compound had a small amount of La3+ doped on the Sr2+ site

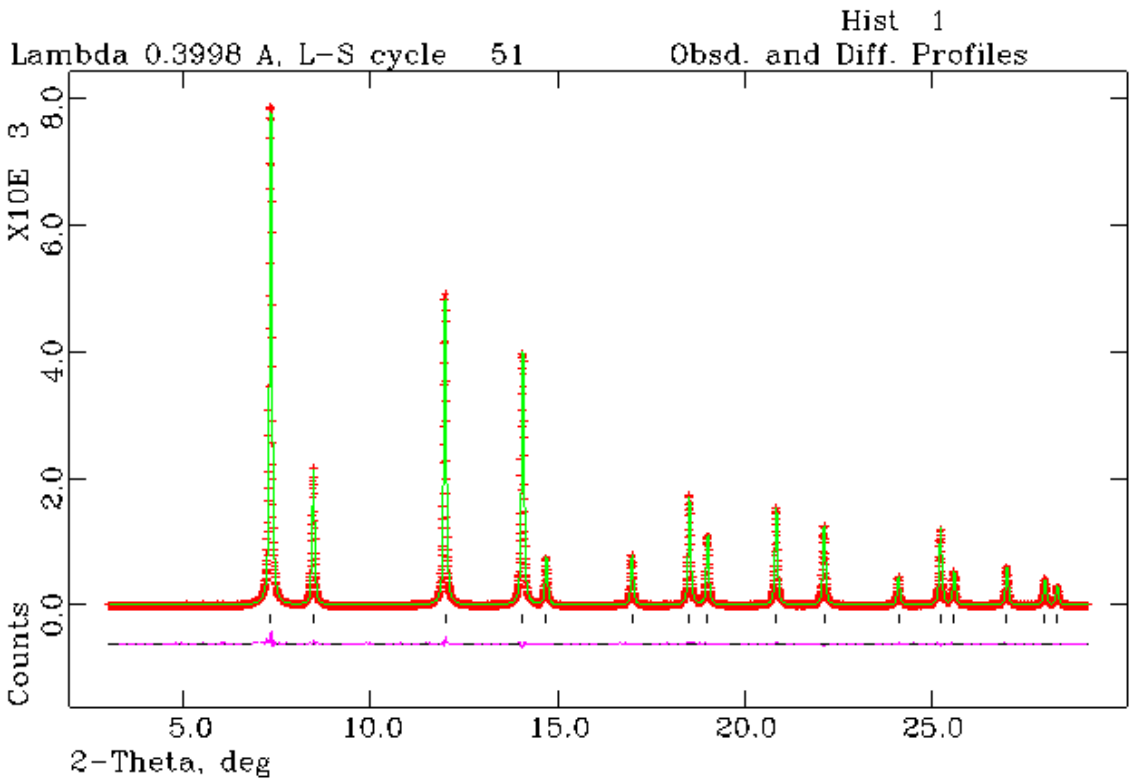
Calibrate the instrument first; leftover broadening is your sample

- Characterize instrument shape parameters using a standard
- Then U, V, W give the integral breadth for Scherrer and Stokes analysis
  - $D_v = K\lambda / (\beta \cos\theta)$
  - $\varepsilon = \beta / (4 \tan\theta)$
  - $(\beta_{obs} - \beta_{inst}) \cos\theta = \lambda / D_v + 4\varepsilon \sin\theta$
- **Williamson-Hall: plot $\beta\cos\theta/\lambda$ vs $4\sin\theta$; slope and intercept give strain and size**



Micronized CeO2 refines to 226 Angstrom domains and 0.011% strain

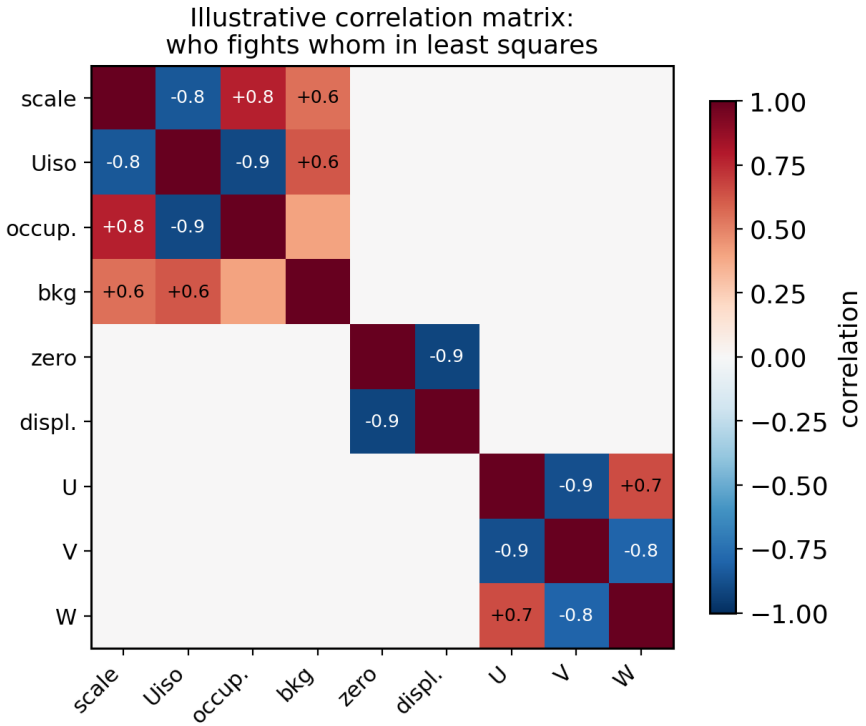




CeO2 before and after micronizing; volume-weighted domain size Dv = 226 A, maximum strain e = 0.011%

Rietveld refinement is a complete quantitative toolkit

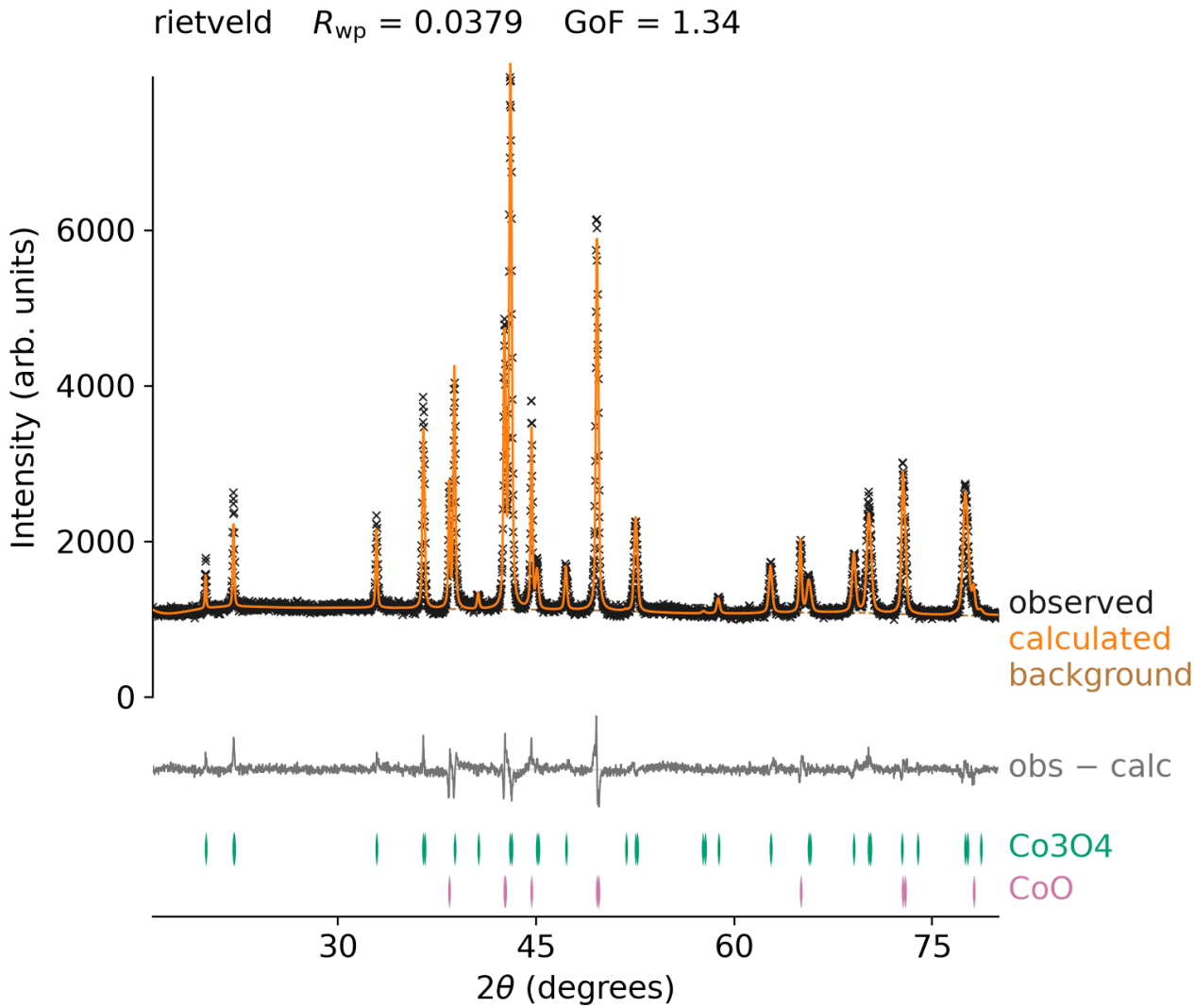
- **Crystal structure information**
  - Atomic positions, thermal displacement parameters, occupancy
- **Beyond the average structure**
  - Preferred orientation, size, strain, quantitative phase analysis (including amorphous material)
- **And remember**
  - Judge fits with the profile plot and chemical sense, not Rwp alone
  - Turn parameters on in a correlation-aware order



A second track: scripted and AI-assisted Rietveld

Same data, same physics, written as a script that you can rerun, diff and share

rietx (rietx.org), cross-checked against GSAS-II on every workshop dataset



Manual and scripted refinement are good at different things

Manual: GSAS-II GUI

you click each parameter on

Strengths

+ Builds intuition: every change shows up in the plot
+ Huge scope: single crystal, magnetic, TOF, rigid bodies
+ Your judgment is in every step

Watch out for

− Hard to reproduce: the protocol lives in your head
− Slow for series and batches
− Easy to accept a fit that looks good (negative B, masked regions)

Scripted: rietx, optionally driven by an AI

the refinement is code

Strengths

+ Reproducible: rerun, diff, share the whole protocol
+ Scales: 12 in-situ scans refined both ways in about 2 minutes
+ Diagnostics check every number and say what not to quote

Watch out for

− Follows bad instructions faithfully (reused another instrument's calibration on request)
− Knows nothing about the specimen: spinning, weighing, milling
− An AI assistant can sound sure and be wrong

Learn and explore with GSAS-II; use a script to reproduce, scale up, and get a second opinion.

A refinement becomes a few lines you can rerun, and every result explains itself

```
import rietx as rx

data = rx.read_pattern("LaB6_Jan2018.raw")
structure = rx.Structure.from_cif("1000055.cif")
instrument = rx.Instrument.bragg_brentano(
    radiation="CuKa", goniometer_radius_mm=250)
instrument.background = rx.auto_background(data)

ref = rx.Refinement(structure, instrument)
result = ref.fit(data, plan="lab_calibrate")
print(result)          # fit + diagnostics
rx.save_instrument_profile(
    ref.fitted_instrument, "lab_profile.json")
```

What the first fit printed

WARNING BISO_NEGATIVE

B = −2.4 Å²: the column is standing in for something else (roughness, absorption, a bending background)

WARNING FLAT_DIRECTION axial S/L ~ H/L

ρ = −1.000: only their combination is measured; fix one

WARNING HIGH_CORRELATION zero ~ displ.

ρ = +0.999 even with the cell held

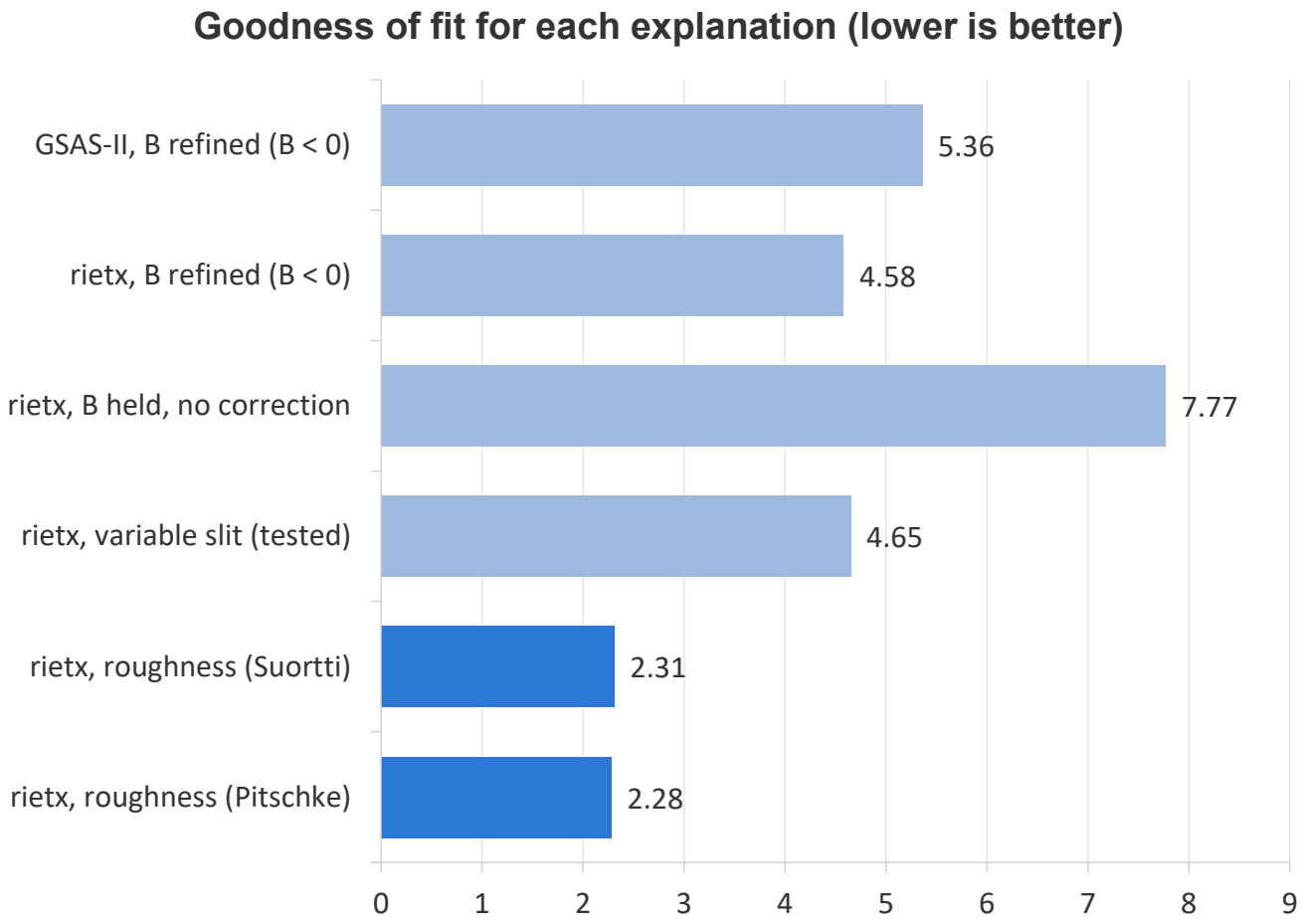
WARNING PATTERN_UNDERSAMPLED

4.3 steps per FWHM: a data-collection limit no refinement can fix

Codes, not adjectives: an AI assistant (or a student) can act on each one, and you can check that it did.

Plans encode the turn-on order: lab_calibrate on a standard, then lab_sample_refine on your specimen.

LaB6: negative B in both codes was really surface roughness



Same held cell (4.15695 Å)	GSAS-II	rietx
Biso La, B (Å²)	−1.49, −0.89	0.3 held
Surface roughness	none	refined
Zero shift (°)	0.023	0.015
Axial (S+H)/L	0.046	0.042
GoF	5.36	2.30
Durbin-Watson	0.17	0.60

The tell: with B held, intensity was 30 % too strong at 21° and 30 to 40 % too weak above 50°, the signature of a rough surface, not of thermal motion.

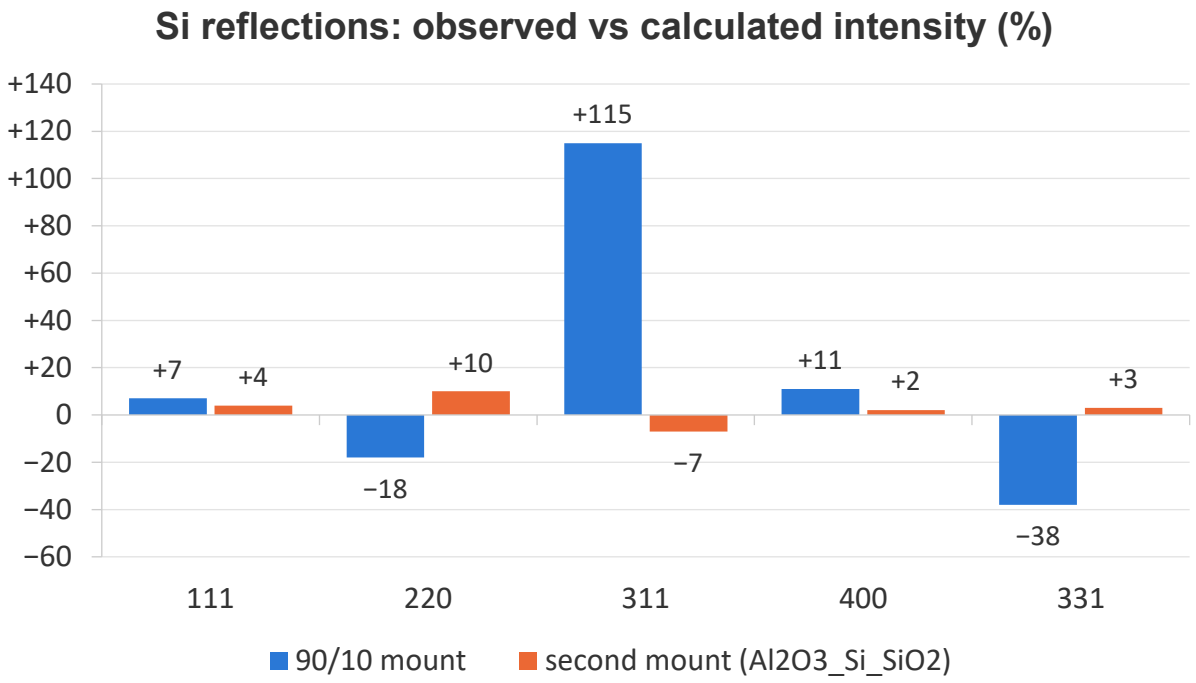
A negative B is a symptom. Test what it is absorbing instead of accepting it, or just holding it.

Al2O3/Si: two codes agree on 3.2 wt% Si; the balance says 10

10.0
wt% Si weighed

3.24
wt% Si, GSAS-II

3.16
wt% Si, rietx



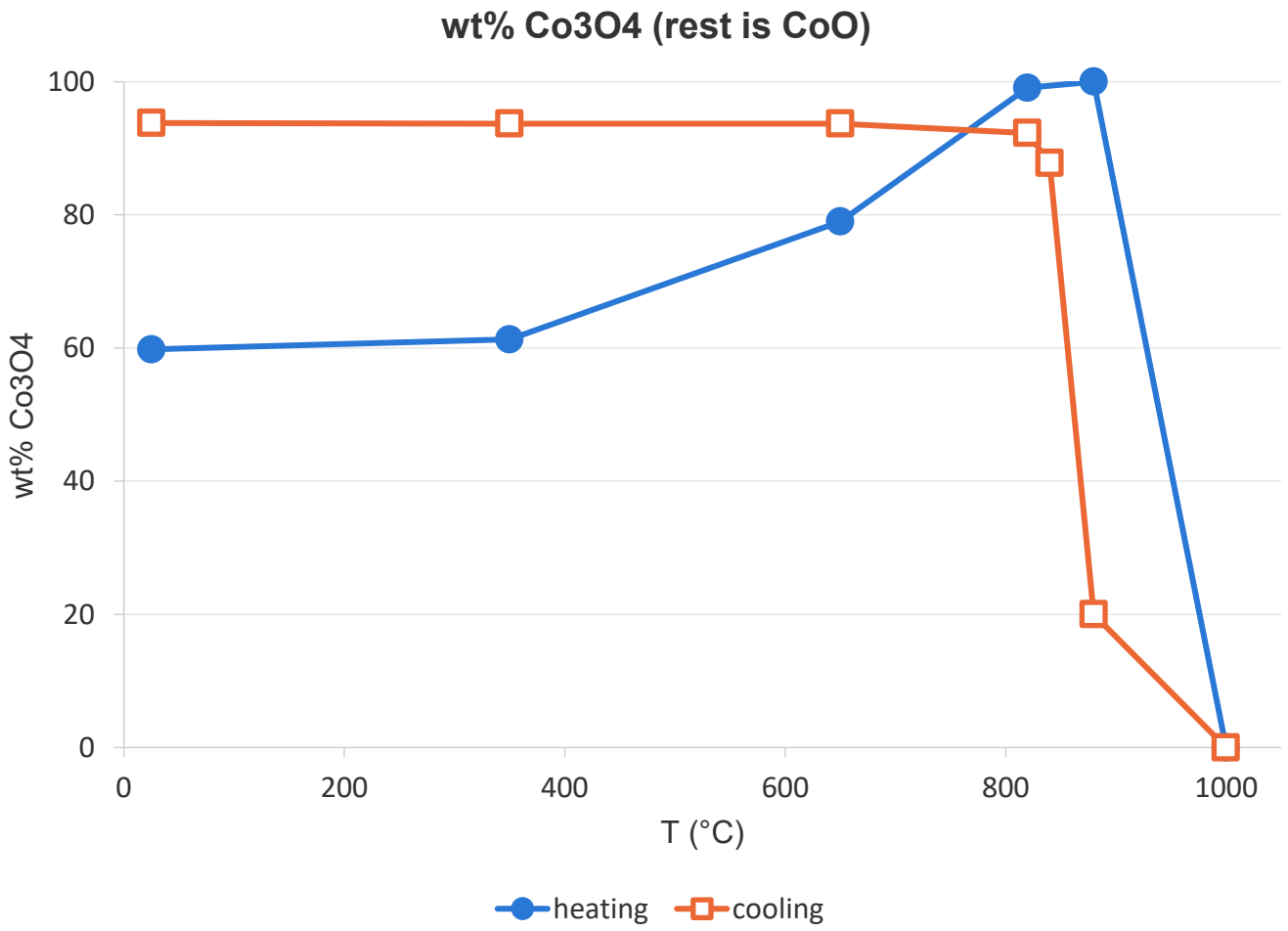
Candidate	Test	Verdict
Amorphous content	internal-standard arithmetic: −217 % amorphous	ruled out
Microabsorption	Brindley, 50 µm Si: 3.2 → 3.8 wt%	too small
Biso choice	held at physical values: 3.6 wt%	**0.5 wt%**
Particle statistics	Si 311 at 2.15×, unspun specimen	**supported**

Two codes agreeing is precision, not accuracy.

Fix it in the lab: spin the specimen, use fine Si, measure several mounts.

The tool did not just report 3.2 wt%; it helped rule causes in and out.

Co3O4: one script refines a 12-scan heating and cooling cycle, and checks itself



What the script found

"Impurity" peaks are Co Kβ
no filter in the beam; modelled, not masked (GoF 2.0 → 1.3 at 25 °C)

Order-independent
chain run forward and backward: no parameter disagrees

Hysteresis
cooling stalls near 94 % (±3 wt% from the minor phase's B)

Matches the older GSAS figure
point for point, plus a first cooling scan at 880 °C

A series is where scripting pays off most: same protocol on every scan, plus checks no one does by hand.

Where the scripted pass caught what my GSAS-II refinements missed

Dataset	My GSAS-II refinement	What the scripted pass found	How it was caught
LaB6 standard	B < 0 on both sites, GoF 5.4	surface roughness; B = 0.3 Å², GoF 2.3	BISO_NEGATIVE, then an intensity-vs-angle trend and rival tests
Al2O3 / Si / SiO2	quartz modelled (0.4 wt%), Uiso ≈ −1.9 Å², GOF 5.5	a broad fluorite-type phase (a = 5.10 Å), most likely ZrO2 from milling; GoF 3.5 → 1.5	add_impurity_phase (confidence 0.9), unmatched broad peaks
Co3O4, 25 °C	5 regions excluded as an impurity	they are Co Kβ lines; modelled, GoF 1.3	positions of Kβ copies of every reflection
Co3O4, 1000 °C	99 % Co3O4, a = 8.230 Å, GOF 7.9	100 % CoO: the spinel is gone (wrong basin)	warm-started series, then an independent cold refit

All four were quick teaching refinements, not final analyses: the kind of fit a busy expert accepts.

The script never gets tired or satisfied: every negative B, unmatched peak and wrong basin gets reported.

It also made mistakes, and only executed numbers caught them

The AI's draft claims that were wrong

“Axial divergence agrees with GSAS-II within 1 %”
at the same held cell: 9 % apart

“The minor phase's B moves fractions < 1 wt%”
measured: 2.5 wt%

“The second mount's Si scatters too”
it sits within 10 %

“The held cell moves the zero by a few thousandths”
0.009°, and χ^2 30 % lower

Blind spots of the tool

Reused a LaB6 calibration from another diffractometer
because it was asked to; only FLOOR_HIT hinted

Its own cold-refit check called a real phase change an artefact
the cold start was outside the fit's reach

Refused an unusual GSAS .prm file
a human had to decide what to drop

Cannot see the sample prep
spinning, weighing, milling media are yours

Treat it like a fast, tireless, overconfident graduate student: check every claim against a number it computed.

Try it: three notebooks on the workshop data, readable on GitHub as is

01

LaB6 instrument calibration

lab_calibrate, a negative-B hunt, and a saved instrument profile

02

Al2O3/Si phase analysis

QPA, GSAS-II side by side, and why 10 wt% reads as 3

03

Co3O4 temperature series

12 scans refined both directions; phase fractions and cells vs T

github.com/sp8rks/Ceramic_Rietveld_Refinement → scripted refinement (rietx)

Run: `pip install -r requirements.txt` then open the notebooks in JupyterLab (Python ≥ 3.11)

With Claude Code: *"Refine LaB6_Jan2018.raw with rietx and explain the diagnostics"* (the rietx skill ships in the repo)

Suggested exercise: refine one dataset in GSAS-II yourself, then run the notebook and compare numbers and diagnostics. Where you disagree, find out who is right.